

Social Security Design and Its Political Support^{*}

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Abstract

The design of social security systems has implications for the political support of intergenerational transfers. We examine the effects of contributive vs. universal pensions, as well as the link between benefits and earnings, on the social security tax rate, labor supply, and capital accumulation. Employing an overlapping generations model featuring heterogeneous households, our analysis aligns with existing results on the effects of retirees' relative political power, ageing, and income inequality. Additionally, we find taxes to be lower when benefits are earnings-related and, if informal retirees are poorer than formal ones and the pension a comparable share of their consumption, higher when pensions are universal. Calibrated to Argentina, the model accounts for about three quarters of the rise in pension spending after the 2005–2010 expansion of coverage. Calibrated to the U.S., France and the UK, ageing is the main determinant of pension spending, the earnings link the second, and income inequality and voting patterns minor. When the electorate also chooses the design of the system, an interior choice requires redistribution through pensions to be costly; with a deadweight cost of flat benefits that rises with the size of the system, calibrated to the U.S., ageing makes benefits more earnings-related, inequality moves the design little, and a single cost parameter orders the designs of the UK, the U.S. and France as observed.

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1 Introduction

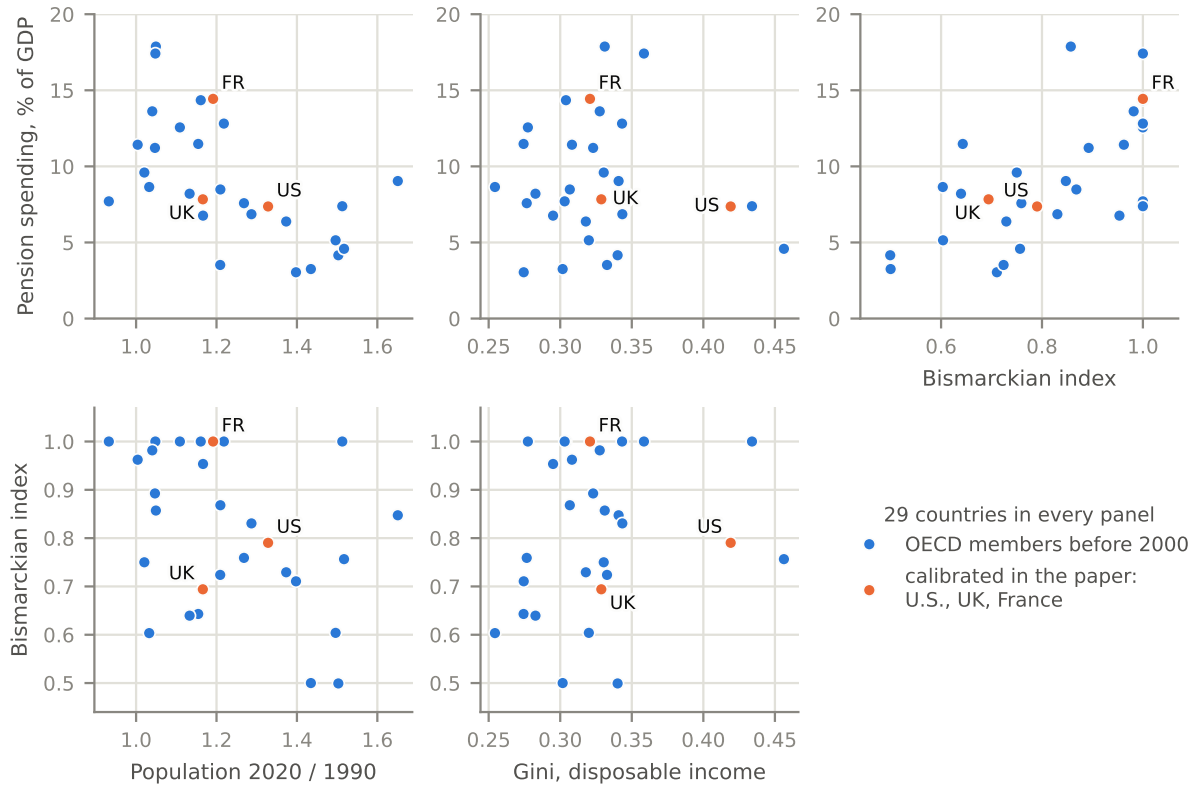
Pensions are the largest transfer programme of most governments. In 2020 public pension spending averaged about 8% of GDP across the OECD and close to 4% in emerging and middle-income economies.¹ Systems differ along two key dimensions. The first concerns eligibility and coverage: some condition benefits on a contribution record, others offer universal entitlements. The second is the link between benefits and past earnings: Bismarckian systems tie the two closely, whereas Beveridgean systems pay flat benefits. Both dimensions shape the labor supply of workers, the redistribution embedded in the system, and the coalition of voters that supports it.

Given their size, it is no surprise that the design of social security systems remains a central concern in policy debates. As populations age, policymakers must navigate difficult trade-offs among intergenerational equity, fiscal sustainability, and labor market efficiency. Moreover, pension arrangements are not determined by actuarial considerations alone: they are the outcome of political processes in which heterogeneous voters and organized interests shape the level and composition of benefits and contributions. A political-economy perspective is therefore critical for understanding why countries differ in the size of their systems and in their design. Figure 1.1 shows how they differ across the 29 countries that were in the OECD before 2000. Pension spending and the Bismarckian–Beveridgean index, the ratio of replacement rates at mean and at half mean earnings, both rise as population growth falls, and they rise together. Neither is related to income inequality. The last fact runs against the leading theory of pension design, that of **CondeRuizP07a1**, in which high inequality leads a coalition of the rich and the poor to choose a small Beveridgean system and low inequality leads the middle class to choose a larger Bismarckian one, and few other politico-economic models have integrated the design. We address this gap with a dynamic politico-economic model and ask whether one political process can deliver both the size of the system and its design, and their relation to ageing but not to inequality.

We study the question in an overlapping generations model with capital accumulation and heterogeneous households. Households consume, save, and supply labor, and as voters, they choose among policies on labor income taxes and retirement benefits. Policy preferences vary by income and age; we model the resolution of these conflicts through probabilistic voting with elections taking place every period. Policy decisions not only impact economic outcomes, but also affect future policy decisions. Voters take these effects into account. We study Markov equilibria in which only fundamental state variables enter. Households differ in their productivity and propensity to vote. In addition, a share of households work in an informal sector or get income from home production,

¹The dispersion is wide even among rich countries: a bit more than 3% of GDP in Ireland and Chile and more than 17% in Italy and Greece (**OECDpen**).

Figure 1.1: Population growth, income inequality, pension spending and design



Note: Sources are the OECD Social Expenditure Database (pension spending), OECD historical population data (population growth), OECD, *Pensions at a Glance 2021*, table 4.1 (Bismarckian–Beveridgean index) and the UNU-WIDER World Income Inequality Database (Gini coefficient); the values by country, the sources and the correlations are in the [online appendix](#), and the construction of the figure is documented in **BergGE26doc**.

where labor is not taxed and savings do not enter the capital stock. Preferences are those of **GreenwoodHH88a1**, so that labor supply depends on the after-tax wage and how current effort raises future benefits. Benefits combine a flat component with an earnings-related one, the weight of the second being the Bismarckian index, and informal households receive a universal pension whose level is the second design parameter.

We show that when utility is logarithmic, politico-economic equilibria retain analytical tractability despite the presence of heterogeneity and institutional richness. With hand-to-mouth informal households, the equilibrium tax is a function of parameters and demographics alone. The analytical frame clarifies how population ageing, retirees' political influence, pension design, and income inequality shape the equilibrium tax rate. A more Bismarckian benefit formula lowers the equilibrium tax, since it shifts benefits from poor retirees, who value them more at the margin, to rich ones, so that retirees as a group support a smaller system. The effect of a more universal pension is ambiguous: it raises the tax when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption. When utility is constant relative risk aversion (CRRA), we show that the state space for equilibrium policies can be reduced to a single

aggregate state and provide numerical solutions.²

We present two empirical applications of the model. First, we study Argentina as the case in which the design changed: from 2005 a series of tax amnesties raised the coverage of the elderly from 68% to over 90%, and pension spending by 1.8% of GDP. Implementing this reform in our calibrated model accounts for about three quarters of the observed rise in pension spending. The model also predicts modest declines in labor supply and savings, consistent with available evidence (**Ruffo**). Second, initially taking the design as given, we calibrate the model to the U.S., France, and the UK, three advanced economies with distinct pension system characteristics, demographics, and inequality profiles. We find that the size of the pension system increases strongly with ageing and flat benefits and responds only little to income inequality, leisure preferences, and voting profiles.

We then let the electorate choose the earnings link as well as the tax. Without a cost of redistribution the choice is a corner. The flat component of benefits is a pure transfer within the retired cohort, from retirees above mean income to those below, and an electorate that weighs retirees by their marginal utility wants all of it: we prove that unless propensities to vote rise sharply with income, the design chosen with the tax is fully Beveridgean, whatever the tax rate, the demography and the income distribution.³

An interior design requires flat benefits to be more expensive than earnings-related ones. We introduce a deadweight cost of the redistribution the flat component performs, read as the tax component of contributions: it is quadratic in the flat share of benefits, proportional to the dispersion of incomes and, per unit of revenue, rising with the size of the system. A single cost parameter is calibrated so that the 2020 U.S. electorate re-elects its own design, and it is tested on the UK and France, whose designs it orders as observed. With this formulation, the model predicts that an ageing population chooses both a more Bismarckian design and higher pension spending, whereas the income distribution has little effect on either; this is in line with the evidence in figure 1.1.

The paper belongs to the literature on politico-economic equilibria of social security (**CooleyS99a1**; **Galasso99a1**; **Forni05a1**; **mkvss08**; **Song12a1**), surveyed by **GalassoP02a1**. As in **mkvss08** and **Song12a1**, under log-GHH preferences the equilibrium has no strategic effects and taxes increase with the political power of retirees, income inequality and ageing; as in **UchidaOa1**, the equilibrium tax is a function of parameters and demographics alone; and as in **mkvss05**, the effect of ageing under CRRA preferences is close to the logarithmic one after recalibration. What we add is a pension system that differs in two dimensions, in coverage and in the earnings link, and workers who differ in productivity and in their propensity to vote. Neither adds a state when

²When informal households can save, the state space includes two states.

³We also explore different ways of choosing the pension design, i.e. choosing the design one period in advance or choosing a permanent structure. We test the variants numerically varying parameters like the labor supply elasticity and the intertemporal elasticity of substitution, but no plausible combination avoids a corner choice for the observed U.S. level of inequality.

informal households are hand-to-mouth, and the calibrated model quantifies what each feature of the design does to the tax rate, labor supply and savings.

On the design of the system, **GalassoP02a1** leave open why some systems are Beveridgean and others Bismarckian, and why the Bismarckian ones are larger, two questions that a vote on the size of the system alone cannot answer. **Casamatta1** study the political support of Bismarckian against Beveridgean benefits, **CondeRuizP07a1** the joint choice of size and design, **KoethenbuergerPP08a1** the trade-off by which more redistributive systems are smaller, and **CremerDMP07a1** the choice of type and generosity. We embed the choice in a dynamic equilibrium with capital accumulation and labor supply, in which the design is chosen with the tax at every date, and take it to the data.

Section 2 presents the model and the equilibrium concepts. Section 3 derives the equilibrium under log-GHH preferences, in which the tax is a function of parameters and demographics alone, and its comparative statics. Section 4 describes how the equilibria are computed outside that case. Section 5 calibrates the model to Argentina and compares its prediction with the observed reform, section 6 calibrates it to the U.S., France and the UK and ranks the determinants of pension spending, and section 7 lets the electorate choose the design. Section 8 concludes. The appendix collects proofs, the derivation of the politico-economic equilibrium, the calibration details and a map of the results across specifications; an online appendix at championape.github.io/SSDclaude holds every table and figure behind sections 5 to 7, the robustness checks and the numerical checks, with a print edition.

2 The Model

2.1 Demographics, Inequality, and Institutions

We consider an economy populated by overlapping generations of workers and retirees. Within each generational cohort, we consider $J + 1$ household types indexed $j = 0, \dots, J$. The $j > 0$ household types allocate time between labor and leisure, pay taxes on labor income, consume, and save. When they reach retirement, they spend the return on their savings and any social security benefits they may receive, and die. Type $j = 0$ households similarly allocate time between labor and leisure, consume, and save for retirement. However, they supply labor to an informal labor market, thus not subject to labor income taxes, and they save for retirement using an informal savings technology. The distribution of type $j > 0$ households is given by γ_j with $\sum_{j>0} \gamma_j = 1$, and the share of type $j = 0$ households is γ_0 . The ratio of workers to retirees in period t follows a deterministic process, and is given by ν_t .

The government runs a pay-as-you-go (PAYG) pension system with a balanced budget. Each period, the government imposes a tax on labor supplied in the formal sector and

transfers the collected amount directly to retirees. We consider social security systems that differ along two dimensions: whether workers need to qualify to receive full benefits or not and the type of benefits provided, either tied to earnings or flat-rate. We describe the different systems in section 2.4.

Policy decisions are taken by a government that acts in the interest of voters, but lacks commitment.

2.2 Technology

In the formal sector, a continuum of competitive firms transform aggregate capital and labor into output. We define the aggregate formal labor supply as

$$H_t = N_t h_t, \quad h_t \equiv \sum_{i>0} \gamma_i \eta_i h_{t,i}, \quad (1a)$$

where N_t is the number of formal workers at time t , $h_{t,i}$ is time allocated to labor by household type i , and η_i is their labor productivity, normalized such that $\sum_{i>0} \gamma_i \eta_i = 1$. Capital is owned by formal retirees and fully depreciates after a period. The economy-wide capital stock per formal worker, k_t , therefore corresponds to the economy-wide per capita savings of formal workers in the previous period, s_{t-1} , normalized by ν_t . We define the aggregate capital stock from

$$K_t = N_{t-1} s_{t-1}, \quad s_{t-1} \equiv \sum_{i>0} \gamma_i s_{t-1,i}. \quad (1b)$$

The representative firm produces output using labor and capital and a conventional Cobb-Douglas technology

$$Y_t = A_t K_t^\alpha H_t^{1-\alpha} \quad (2)$$

In equilibrium this implies the following factor prices:

$$R_t = \alpha A_t \left(\frac{s_{t-1}}{\nu_t h_t} \right)^{\alpha-1}, \quad w_t = (1 - \alpha) A_t \left(\frac{s_{t-1}}{\nu_t h_t} \right)^\alpha. \quad (3)$$

Informal households produce with a linear technology, so the informal wage w_t^0 and the return on informal savings R_t^0 are proportional to the formal sector's factor prices.

2.3 Preferences and Household Choices

Households value consumption and leisure, and discount the future at factor $\beta \in (0, 1)$. Representing utility by function $u(c, h)$ with $u'_c \geq 0, u'_h \leq 0, u''_{cc} \leq 0, u''_{hh} \geq 0$, the welfare

of a young household of type i born in period t is defined by

$$u(c_{1,t}^i, h_{t,i}) + \beta u(c_{2,t+1}^i, 0). \quad (4)$$

The budget constraints for formal and informal workers are given by:

$$c_{1,t}^i = w_t \eta_i h_{t,i} (1 - \tau_t) - s_t^i, \quad i > 0, \quad (5a)$$

$$c_{1,t}^0 = w_t^0 \eta_0 h_{t,0} - s_t^0, \quad (5b)$$

where τ_t is the social security tax rate. As retirees, households receive a pension b_{t+1}^i that may depend on the individual household's contribution, and thus on their labor supply, h_t^i :

$$c_{2,t+1}^i = s_t^i R_{t+1} + b_{t+1}^i(h_{t,i}), \quad i > 0, \quad (5c)$$

$$c_{2,t+1}^0 = s_t^0 R_{t+1}^0 + b_{t+1}^0(h_{t,0}). \quad (5d)$$

Optimization yields the first order conditions that represent a leisure/consumption tradeoff and a conventional Euler condition for intertemporal consumption smoothing:⁴

$$-\frac{\partial u_i}{\partial h_{t,i}} = \frac{\partial u_i}{\partial c_{1,t}^i} \left(w_t (1 - \tau_t) \eta_i + \frac{\partial b_{t+1}^i / \partial h_{t,i}}{R_{t+1}} \right), \quad i > 0 \quad (6a)$$

$$-\frac{\partial u_i}{\partial h_{t,0}} = \frac{\partial u_i}{\partial c_{0,t}^i} \left(w_t^0 \eta_0 + \frac{\partial b_{t+1}^0 / \partial h_{t,0}}{R_{t+1}^0} \right), \quad (6b)$$

$$\frac{\partial u_i}{\partial c_{1,t}^i} = \beta R_{t+1} \frac{\partial u_i}{\partial c_{2,t+1}^i}, \quad i > 0 \quad (6c)$$

$$\frac{\partial u_i}{\partial c_{1,t}^0} = \beta R_{t+1}^0 \frac{\partial u_i}{\partial c_{2,t+1}^0}. \quad (6d)$$

Optimal savings and labor supply decisions can, in general, be written as

$$h_t^i = h_t^i(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}) \quad (7a)$$

$$s_t^i = s_t^i(s_{t-1}, s_t, h_t, h_{t+1}, \tau_t, \tau_{t+1}), \quad (7b)$$

where the dependence on h_{t+1}, τ_{t+1} comes from the pension system.

2.4 Social Security System

The government runs a pay-as-you-go pension system with a balanced budget. Every period, the government taxes labor income in the formal sector and pays pensions to current retirees.

⁴We are using shorthand notation $u_i \equiv u(c_i, h_i)$ with timing implicit by the variable we take derivatives with respect to.

We consider systems that differ along two dimensions. First, in *contributive* systems workers have to meet a contribution requirement to receive full pension benefits. In contrast, in a *universal* system all workers are eligible to receive pensions upon retirement, regardless of whether they contributed to it or not.

Second, pension schemes also differ according to the relationship between the level of contributions and benefits. *Bismarckian* systems provide earnings-related benefits; this gives workers an additional incentive to work as they perceive this increases their future pensions. In contrast, *Beveridgean* systems offer flat payments and lack this incentive. We model this by making benefits to consist of a common amount for all formal workers and a second component that is proportional to past labor income.⁵

Benefits are thus given by:

$$b_t^i = [\theta h_{t-1,i} \eta_i + (1 - \theta) h_{t-1}] \bar{b}_t \quad (8a)$$

$$b_t^0 = \epsilon h_{t-1} \bar{b}_t \quad (8b)$$

where ϵ determines the level of universal pensions received by informal workers, θ measures how Bismarckian are the benefits of formal workers, and $\bar{b}_t$ is the average formal worker's pension benefits. With $\epsilon = 0$ the system is purely contributive and there are no pensions for informal workers. With $\epsilon = 1 - \theta$ they receive the basic common pension of formal workers, and with $\epsilon = 1 - \theta + \theta \eta_1 h_{t-1,1} / h_{t-1}$ the system is purely universal in the sense that informal workers receive the same pensions as the type of formal working household with lowest productivity.

The balanced budget requirement implies that

$$\nu_t w_t \tau_t h_t = \sum_{j \geq 0} \gamma_j b_t^j$$

This implies that the average pension rate $\bar{b}_t$ is:

$$\bar{b}_t = \frac{\nu_t w_t h_t \tau_t}{h_{t-1} (1 + \gamma_0 \epsilon)}. \quad (8c)$$

2.5 Elections

Elections take place at the beginning of each period. We assume that preferences are aggregated through probabilistic voting.⁶ Thus, policy maximizes a convex combination of the objective functions of all groups of voters, where the weights reflect the groups' sizes and their responsiveness to policy changes. We allow for age related variation in

⁵This formulation has been frequently used in the study of redistributive social security, see e.g. Casamatta1.

⁶See LindbeckW87a1.

responsiveness, reflected in a per capita political influence weight of unity for young voters and a per capita weight of $\omega \geq 0$ for retired voters.

2.6 Equilibrium

Competitive Equilibrium. In economic equilibrium, firms and households optimize given factor prices, policies, and a vector of aggregate states z_t , which includes exogenous demographics as well as the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}_j$. In a rational expectations equilibrium, the path of policies $\{\tau_s\}_{t \leq s \leq T}$ fully determines an allocation and price system.

Conditional on some initial states z_0 and a policy sequence $\{\tau_t\}_{0 \leq t \leq T}$, a competitive equilibrium is given by an allocation and price system such that

- i. households optimize;
- ii. aggregate states are defined according to (1), markets clear, and factor prices are determined according to (3) for all t ; and
- iii. the government budget constraints in (8) are satisfied for all t .

Politico-economic equilibrium. In politico-economic equilibrium political decision makers optimally choose tax rates, taking all implications of their actions into account and forming rational expectations about future policy choices. We assume that these choices are Markov, i.e. they are functions of the fundamental state variables. The decision maker at date t takes the distribution of states $\{s_{t-1,j}, h_{t-1,j}\}$ as well as the continuation policy $\tau_t^{+1}(z_t)$ as given. Furthermore, given the continuation tax function, the policymaker takes as given the following law of motion for the state variables as a function of current policy choices:

$$z_{t+1} = \zeta_t(z_t, \tau_t, \tau_t^{+1}). \quad (9)$$

Subject to (3), (8), (9), the policymaker chooses τ_t to maximize

$$\mathcal{W}_t(z_t, \tau_t; \tau_t^{+1}(z_{t+1})) \equiv \sum_j \gamma_j \mu_j \left[\omega \mathcal{O}_j(z_t, \tau_t) + \nu_t \mathcal{Y}_j(z_t, \tau_t; \tau_t^{+1}(z_{t+1})) \right],$$

where the objective function is the weighted sum of indirect utilities from retirees $\mathcal{O}_j$ and workers $\mathcal{Y}_j$, with weights reflecting relative population sizes through ν_t and γ_j , relative political weight of retirees ω , and propensity to vote μ_j .

Definition 1. A politico-economic equilibrium as of period t conditional on z_t consists of a sequence of tax functions, $\{\tau_t(\cdot)\}_{t \geq t}$; a sequence of continuation tax functions,

$\{\tau_\iota^{+1}(\cdot)\}_{\iota \geq t}$; a sequence of laws of motion for the state variables, $\{\zeta_\iota(\cdot)\}_{\iota \geq t}$; policy choices, $\{\tau_\iota^*\}_{\iota \geq t}$; and a competitive equilibrium allocation such that

i. tax functions are optimal subject to continuation tax functions:

$$\tau_\iota(z_\iota) \in \arg \max_{\tau_\iota} \mathcal{W}_\iota(z_\iota, \tau_\iota; \tau_\iota^{+1}(\cdot)) \text{ for all } z_\iota, \text{ s.t. (3), (8), (9), } \iota \geq t;$$

ii. continuation tax functions are consistent with tax functions:⁷

$$\tau_{\iota-1}^{+1}(z_\iota) = \tau_\iota(z_\iota, \tau_\iota^{+1}(z_{\iota+1}(\cdot))) \text{ for all } z_\iota, \iota \geq t;$$

iii. laws of motion are consistent with the policy and continuation policy functions according to (9);

iv. equilibrium tax choices are generated by the continuation tax function,

$$\{\tau_\iota^*\}_{\iota \geq t} = \tau_{\iota-1}^{+1}(z_\iota),$$

and $\{\tau_\iota^*\}_{\iota \geq t}$ implements a competitive equilibrium allocation.

For infinite horizon and a recursive time-autonomous structure, the policy and continuation policy functions are time-autonomous functions of the state as well, and conditions i. and ii. above are combined in a fixed point requirement.

3 Logarithmic-GHH Preferences

This section derives the model's analytical results. We follow **GreenwoodHH88a1** and assume

$$u(c_{1,t}^i, h_{t,i}) = \ln(\tilde{c}_{1,t}^i), \quad \tilde{c}_{1,t}^i \equiv c_{1,t}^i - \frac{X_i \xi}{1 + \xi} (h_{t,i})^{\frac{1+\xi}{\xi}} \quad (10)$$

where ξ is the Frisch elasticity of labor supply and X_i is a type-specific taste for leisure. Proposition 1 characterizes the economic equilibrium under these log-GHH preferences. If informal households are in addition hand-to-mouth, the politico-economic equilibrium carries no state (proposition 2) and the tax rate is a function of parameters and demographics alone. The quantitative sections keep log-GHH preferences as the benchmark and let informal households save for retirement.

⁷With a finite horizon, $\tau^{T+1}(z_T) = \emptyset$.

3.1 Economic Equilibrium

With log-GHH preferences, formal households i supply labor and save according to

$$h_{t,i} = \left(\frac{\eta_i w_t (1 - \tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i} \right)^\xi \quad (11a)$$

$$s_{t,i} = \frac{\beta}{1 + \beta} \left[w_t (1 - \tau_t) h_{t,i} \eta_i - \frac{X_i \xi}{1 + \xi} (h_{t,i})^{\frac{1+\xi}{\xi}} \right] - \frac{b_{t+1}^i / R_{t+1}}{1 + \beta}. \quad (11b)$$

Appendix A shows that in economic equilibrium every relevant variable x (savings, labor supply and consumption) is a power of capital per worker times a coefficient,

$$x_t = \Theta_{x,t} \left(\frac{s_{t-1}}{\nu_t} \right)^{\sigma_x},$$

where the coefficient $\Theta_{x,t}$ depends on parameters and taxes and the exponent σ_x on parameters only.

Proposition 1. *In economic equilibrium we have the following:*

- i. *Aggregate labor is decreasing in τ_t , ϵ and γ_0 and increasing in τ_{t+1} , θ . Aggregate savings are decreasing in τ_t , τ_{t+1} , and θ and are increasing in ϵ and γ_0 .*
- ii. *Labor income inequality and relative labor supply are independent of pension policy and given by*

$$\frac{\eta_i h_{t,i}}{h_t} = \frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_h}, \quad \Gamma_h \equiv \sum_i \frac{\gamma_i \eta_i^{1+\xi}}{X_i^\xi}. \quad (12)$$

- iii. *Savings of formal households relative to average savings are independent of τ_t and linear in τ_{t+1} :*

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{\Gamma_h} + \frac{1 - \alpha}{\alpha} \frac{\tau_{t+1}}{1 + \gamma_0 \epsilon} \frac{1 - \theta}{1 + \beta} \left(\frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_h} - 1 \right). \quad (13)$$

- iv. *When $\tau_{t+1} = 0$ or with fully Bismarckian pensions ($\theta = 1$), the heterogeneity in savings exactly matches labor income inequality. An increase in τ_{t+1} or a decrease in θ amplifies the spread in private savings.*
- v. *Informal savings increase in τ_t and θ and decrease in τ_{t+1} and ϵ ; current taxes also raise their ratio to formal savings.*

Part i of proposition 1 states that current taxes reduce labor supply and savings, while future taxes increase labor supply, since higher expected benefits make work more

attractive, and reduce the need to save. A higher θ increases labor supply for the same reason; with GHH preferences this raises the marginal utility of $\tilde{c}_{1,t}$ and so, through the Euler equation, lowers aggregate savings. A higher ϵ shifts future benefits toward informal households and lowers the benefit formal households expect, so that they save more and work less, and a larger share of informal households has the same effects.

Parts ii and iii reduce the state space of the politico-economic equilibrium, as the next subsection shows. Part iv implies that higher future taxes or a lower θ , when pensions are redistributive, widen the dispersion of savings and thus of private wealth. Part v works through the universal pension, since informal earnings do not respond to pension policy. The pension's present value to an informal household is proportional to formal savings, with a factor that rises with future taxes and ϵ : current taxes and a higher θ , which lower formal savings, therefore raise informal savings, while future taxes and a higher ϵ raise the present value of the pension and lower them.

3.2 Politico-Economic Equilibrium

In a Markov equilibrium the distributions of past savings and past labor supply are in principle state variables of the political choice of the tax rate. Part ii of proposition 1 removes the distribution of labor supply, since taxes do not move relative labor supply.⁸ If informal households save, their past savings are a further state, and one that both current and future taxes move. For the analytical results we therefore take informal households to be hand-to-mouth: they do not save, so that their Euler condition (6d) does not apply, and when old they receive an endowment tied to the informal wage, which can be read as work in old age at a lower productivity.⁹

With log-GHH preferences and hand-to-mouth informal households, the policymaker maximizes

$$\begin{aligned} \mathcal{W}_t(z_t, \tau_t; \tau_t^{+1}(z_{t+1})) = & \sum_i \gamma_i \mu_i \left[\omega \ln \left(c_{2,t}^i \right) + \nu_t \left(\ln \left(\tilde{c}_{1,t}^i \right) + \beta \ln \left(c_{2,t+1}^i \right) \right) \right] \\ & + \gamma_0 \mu_0 \left[\omega \ln \left(\tilde{c}_{2,t}^0 \right) + \nu_t \left(\ln \left(\tilde{c}_{1,t}^0 \right) + \beta \ln \left(c_{2,t+1}^0 \right) \right) \right] \end{aligned} \quad (14)$$

Proposition 2. *In politico-economic equilibrium with hand-to-mouth informal households, all consumption functions are log-separable in the endogenous state variables, making policy only a function of parameters, including system characteristics and demographics. Politico-economic equilibrium taxes are increasing in ω , decreasing in ν_t , and increasing in income inequality. Furthermore they are decreasing in θ . The effect of ϵ is ambiguous and depends on the pension's share of consumption of informal and formal*

⁸This property of GHH preferences also holds under CRRA preferences (appendix A.1).

⁹We model old-age income as an endowment of goods to keep the model as symmetric with the general case as possible.

retirees.

The economic equilibrium is log-separable in the aggregate states (appendix A.2), and the one remaining vector of states, the distribution of formal households' savings $s_{t-1,i}/s_{t-1}$, is a function of parameters and the current tax alone; by the same token, the distribution carried into the next period does not depend on the current choice (appendix B.1). Despite two dimensions of pension design and heterogeneous households, the politico-economic equilibrium therefore shares the property of **mkvss08**; **ssvsde15**; **Song12a1**; **UchidaOa1**, among others, that policy is a function of parameters and demographics only, although the equilibrium conditions are more complex and in most cases can be solved only numerically.

Notwithstanding the orthogonality of current policy with state variables, the trade-offs underlying the equilibrium tax rates are dynamic in nature as they relate contemporaneous policy decisions to capital accumulation and thus future wages and tax revenue. The tractability of the model comes from specifying functional forms that render the factor price elasticities and the derivatives of the indirect utility functions orthogonal to the capital stock and relative labor supply and relative savings decisions independent of current taxes.

The comparative statics in ω and ν_t mirror those in **mkvss08** and **Song12a1**, and the latter also shows that the equilibrium tax rate increases with income inequality. A distinct contribution to the politico-economic literature on social security is the prediction that a more Bismarckian benefit formula weakens political support for taxation. The reason for this is that a higher θ shifts benefits from low-income to high-income retirees, and the former, with their higher marginal utility, weigh more in the political objective, so that retirees' support for taxes declines in the aggregate. A higher ϵ shifts benefits from formal to informal retirees. With log utility a retiree's support for the tax is the pension's share of their consumption, so the shift raises the support of informal retirees and lowers that of formal ones (appendix B.1 gives the exact condition for the sign). Taxes rise when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption, as in Argentina (section 5), because their smaller pension then rises in a larger proportion than formal pensions fall.

When informal households save or preferences are CRRA, the tax rate depends on the state and the equilibrium has to be solved numerically; section 4 describes how.

4 Numerical Methods

This section describes how the politico-economic equilibria of sections 5–7 are computed, with the algorithms and the code in the technical documentation (**BergGE26doc**). The quantitative politico-economic literature computes a time-invariant policy function

([KrusellRR99a1](#); [Forni05a1](#); [Song12a1](#)) or compares steady states across demographic scenarios ([Galasso99a1](#)). Along a demographic transition, however, the Markov policy function is date-specific, since the policymaker at t faces its own mix of cohorts, ν_t , and a continuation policy set further along the transition. The distinction is immaterial in the equilibrium of [proposition 2](#), which carries no state, but not under CRRA preferences or with informal savers, where the continuation reaches the tax through the state, or in the design choice, where a design chosen at t is weighed against the electorate of $t + 1$.¹⁰ We therefore compute date-specific policy functions by backward induction from a date T ([appendix C](#)), the end of a finite horizon or the start of a steady state in an infinite one. For a large T the two agree away from T . We report the finite-horizon solution, which approximates the Markov equilibrium in the limit of finite-horizon economies, the class in which [mkvss08](#) establish uniqueness.

In the general case with CRRA preferences, the distribution of labor supply remains independent of policy (part ii of [proposition 1](#)) and the distribution of past savings is pinned down by the current tax together with past aggregate savings ([appendix A.1](#)). While the distribution of past savings is indeed relevant for the political problem, we exploit that it only depends on the current tax and past aggregate savings ([appendix A.1](#)) to eliminate it from the state space, keeping only the level of past savings s_{t-1} : we work with the first order condition in τ_t derived with the distribution of past savings held fixed, and then replacing it with the one consistent distribution for each candidate tax and for each state s_{t-1} . Informal savers add the ratio $s_{t-1,0}/s_{t-1}$ of informal to formal savings to the state space, which also depends on τ_{t-1} ([appendix B.2](#)).

When the policy function features one or more states, the current tax moves the states and affects future policy choices. We measure the marginal effect by numerical differentiation of the $t + 1$ solution grid, which the backward induction has already computed. A first order condition can have several roots, some of them minima, and the optimum can lie at a corner where it has none, so we evaluate the condition on a grid of taxes and test the corners and all its roots. A candidate passes if it maximizes the political objective with its own consistent distribution held fixed.¹¹

Each reported path starts from the steady state under the first period's policy. Parameters whose targets are equilibrium outcomes, such as the discount factor and the political weight of the old, are calibrated by re-solving the equilibrium path at each trial value until the targets of [sections 5](#) and [6](#) hold in the calibration year.

Under log preferences the choice of the design in [section 7](#) adds no state either. The

¹⁰In our applications, the distinction has a modest effect on results: a time-invariant policy function, solved at each date's own ν_t and applied along the projected path, misses the tax rate by roughly 0.2 p.p., and by up to 0.5 p.p. in the Argentine calibration at the lowest intertemporal elasticity, and misplaces the design by up to 0.059.

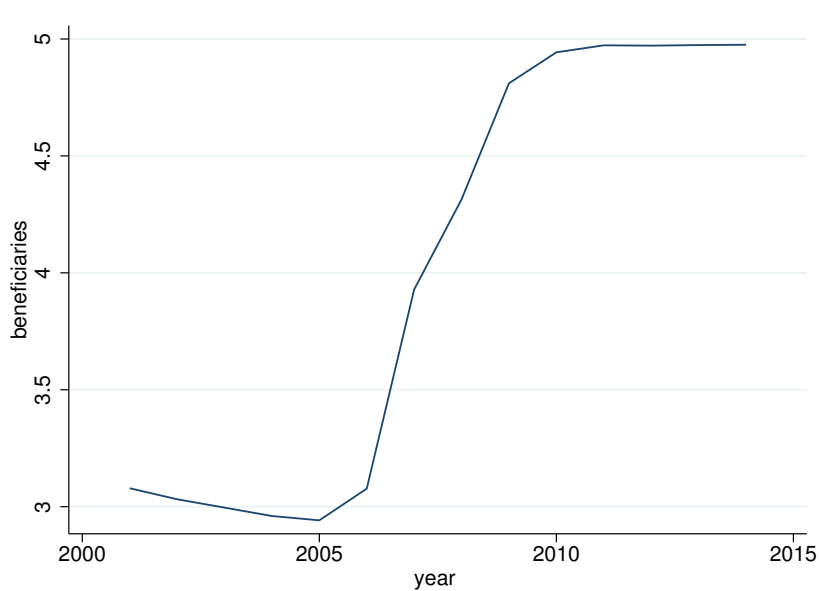
¹¹Several passing candidates would be ranked by payoff, an assumption about equilibrium selection that none of our reported equilibria requires: exactly one passes at every state and date of every solve that records the count ([online appendix](#)).

design chosen at t applies from $t + 1$ and depends neither on the design chosen in the previous period, which sets current benefits, nor on aggregate savings, and the tax at each candidate design is found as above, without a state. Each date's choice is therefore a search over a grid of designs. Under CRRA preferences neither property survives: the equilibrium is a backward recursion over policy functions of the state (s_{t-1}, θ_t) .

5 Pension System Reform in Argentina

Argentina is a middle-income country with high labor informality. In 1994 the social security system was reformed tying full benefits to 30 years of contributions, which left about a third of the elderly without a pension by 2004.¹² From 2005 a series of tax amnesties let workers with shorter contribution records draw a pension. They added 2.7 million beneficiaries by 2011 (**RoffmanA15a1**), raising the number of pensioners from about 3 to 5 million (figure 5.1) and the coverage of the elderly from 68% to over 90% (**Roffman**). Thus, in a few years, Argentina experienced a significant increase in non-contributive pensions: Between 2005 and 2010 social security spending increased by 1.9% of GDP with most of this increase, 1.8% of GDP, due to the tax amnesties (**CetrangoloG**).

Figure 5.1: Pension coverage in Argentina



Note: Beneficiaries of the national pension system (SIPA) and of non-contributive old-age pensions, annual averages in millions. Source: **CetrangoloG**, from the Boletín Estadístico de la Seguridad Social.

The following outlines the calibration procedure and presents the main results from the pension reform. Appendix D and the [online appendix](#) keep calibration details and

¹²The reform also let workers opt for a fully funded system, whose funds were nationalized in December 2008.

supplementary evidence from the reform.¹³

Calibration. We calibrate the model with informal savers to Argentina before the reforms and read the tax amnesties as an increase in ϵ . Informal households work when young and save for retirement, facing the same rate of return on savings as formal households. A model period is thirty years, the calibration year is 2010, and every household has the same propensity to vote. Table 1 reports the result. The pre-reform $\epsilon = 0.21$ is the basic pension of retirees without a full contribution record, relative to the benefit of the least productive formal workers and discounted because it starts five to ten years after the formal retirement age, and $\gamma_0 = 0.47$, informal households per formal household, puts informal households at 32% of all, the share of the population over 65 without a pension before the reform (**Roffman**). The Bismarckian index $\theta = 0.84$ matches the observed ratio of replacement rates at mean and half-mean income. The reform raises ϵ to the universal level $\epsilon^U = 1 - \theta + \theta\eta_1 h_{t-1,1}/h_{t-1}$ of section 2, at which informal retirees receive the benefit of the least productive formal ones. We impose a formal-sector capital share of $\alpha = 0.35$, between the values implied by the survey's informal earnings and by shadow-economy estimates of informal output, a Frisch elasticity of $\xi = 0.30$, and a ratio of workers to retirees ν_t from thirty-year gross population growth in census data and projections, as in **mkvss08**.

The remaining parameters are identified jointly with the equilibrium path. The formal productivities η_i match the income distribution across the four formal quartiles of the survey, with a taste for leisure X common to them whose level is pinned by the observed formal workweek of 42.5 hours.¹⁴ Relative hours across quartiles are then a prediction, which reproduces the survey's ranking and overstates the hours of the top quartile, 1.18 against 1.10 times the formal average. Four parameters are solved against four targets in 2010: β against a capital-output ratio of 3.23, ω against an equivalent social security tax rate of 10.9%, which is pension spending of 7.1% of GDP before the 2001 crisis divided by the labor share of formal output, and η_0 and X_0 against the informal households' income and hours relative to the formal average.

The reform. Table 2 reports the effect in 2010 of an unanticipated and permanent rise of ϵ to ϵ^U under log-GHH preferences, first with the tax rate held at its pre-reform path and then chosen politically. We find that taxes increase, which drives down savings and labor supply; with taxes held fixed, formal savings rise, since informal households now hold a claim on the pension system and save less, and hours fall only marginally.

¹³Appendix D holds the data, the identifying formulas and the sources of the calibration. The online appendix holds the reform along the horizon and across the intertemporal elasticity of substitution, the equilibrium over the whole range of designs, and the calibration under the alternative treatment of the taste for leisure.

¹⁴Identifying one X_i per quartile from the survey's relative hours instead makes hours data and leaves their average level unidentified (appendix D.5). Since X enters no aggregate, only η_i and X_i differ, and every result of this section is the same under both.

Table 1: Calibration, Argentina

Parameter	Value	Target
ϵ (<i>pre-reform</i>)	0.21	Basic pension, 70% of the minimum, discounted
θ	0.84	Replacement rate dispersion
α	0.35	Factor income shares
ν_t	[1.65, 0.96]	30-year gross population growth rates
ξ	0.30	Elasticity of labor supply
β	0.65	Capital–output ratio of 3.23
X	1.94	Average formal workweek of 42.5 hours
η_i	[0.64, 0.90, 1.14, 1.87]	Income distribution
γ_0	0.47	Elderly without a pension, 32% of all households
ω	1.45	Social security tax of 10.9%
η_0	0.344	Informal relative income, eq. (D.1)
X_0	0.804	Informal relative working hours

Note: Our default specification relies on $\rho = 1$.

In 2010 the model predicts that the social security tax rate rises by 2.1 p.p., which at the labor share of formal output is pension spending of 1.4% of GDP, about three quarters of the 1.8% of GDP that the amnesties added. The reason for this is that the reform moves benefits toward informal retirees, who are poorer than formal ones and value benefits more at the margin.¹⁵ The savings rate falls by 0.4 p.p. of GDP and the average workweek by a quarter of an hour, the fall in savings in line with the estimate of **Ruffo** for the amnesty. The tax response builds over time, to 3.5 p.p. above the pre-reform path by 2040, because informal households, now covered, save less for retirement and as retirees demand more taxation.

Table 2: Pension system reform, year 2010

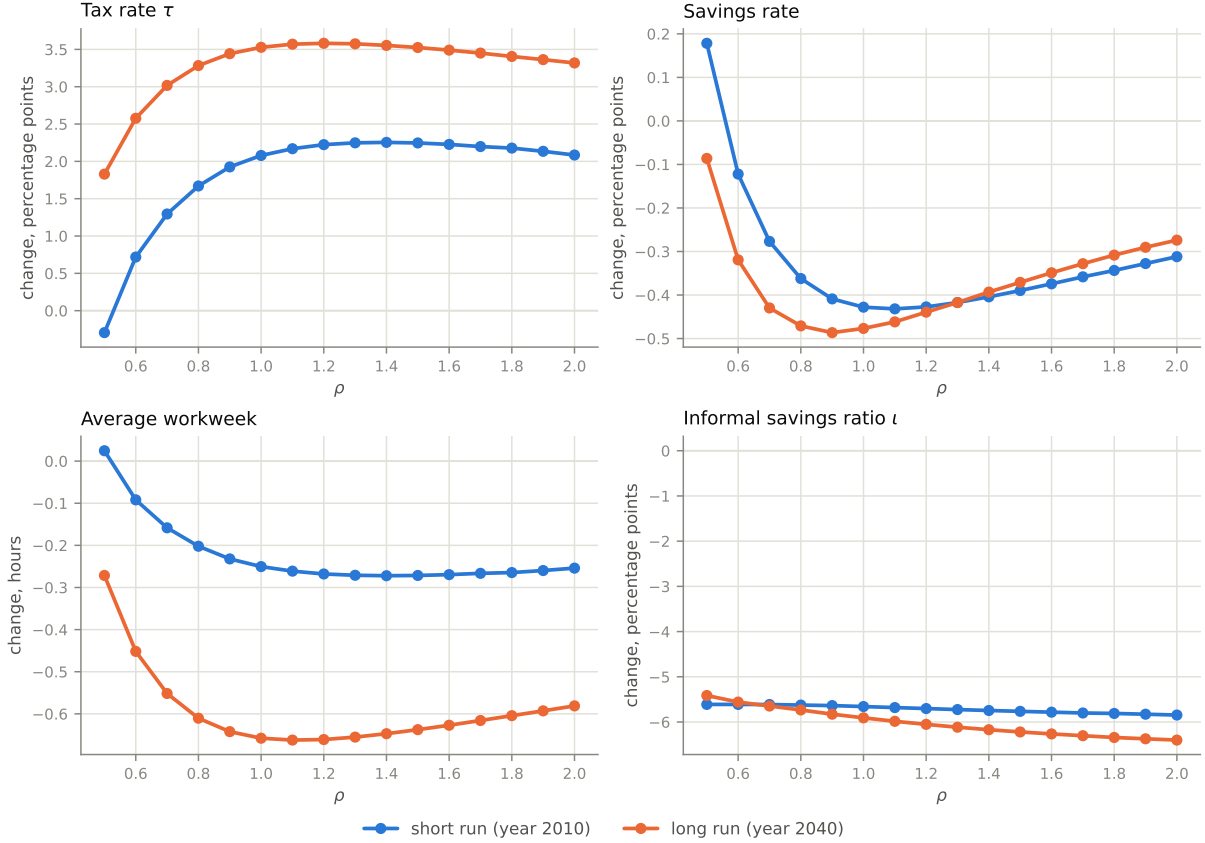
Scenario	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	10.92%	14.40%	42.54
Economic Equilibrium	10.92%	14.73%	42.46
Full effect	13.00%	13.97%	42.29

Note: The reform permanently shifts $\epsilon = 1 - \theta + \theta\eta_1 h_1/h$ at the calibration year, unanticipated. The “Economic Equilibrium” scenario solves for the new economic equilibrium path with taxes as in the Baseline scenario, but with the new permanent ϵ ; the “Full effect” scenario lets taxes be determined by the politico-economic equilibrium. $\rho = 1$. The savings rate is savings relative to GDP.

¹⁵The tax also rises when informal households are hand-to-mouth, consistent with condition (B.5) in appendix B.1.

The effect of the intertemporal elasticity of substitution. Since the environment has no uncertainty, we read the parameter ρ of CRRA preferences as the intertemporal elasticity of substitution (IES). The evidence on its size is inconclusive,¹⁶ and we keep logarithmic utility as the benchmark. Figure 5.2 repeats the reform with the pre-reform economy calibrated at each ρ and traces the effects in 2010 and 2040 for taxes, savings rates, average workweek, and informal savings.

Figure 5.2: Short- and long-run effects of the pension reform as a function of ρ



Note: Each panel is the change against the pre-reform path of the same ρ . The savings rate is savings relative to GDP.

The effects of the pension reform are nonlinear in ρ , in particular for values lower than one. When $\rho < 1$ the effects of the pension reform are dampened across all metrics, and at the lowest values of the IES the tax rate even falls slightly on impact. The reason for this is that with a low IES, the young resist taxation more strongly, so the recalibrated economy needs a larger political weight of the old to reproduce the observed tax rate and responds less to a change in benefits. For an IES between 1 and 2 the results stay close to the log benchmark: the tax rises by 2.1 to 2.3 p.p. in 2010 (3.3-3.6 p.p. by 2040), the workweek shortens by roughly 0.25 hours in 2010 (0.6-0.7 hours by 2040), and the fall in

¹⁶Estimates from aggregate consumption data are low, which **Güvenen et al.** reconciles with higher values from output and wealth data through limited asset market participation, and asset-pricing models with Epstein–Zin preferences and long-run or disaster risk require an IES above one (**Bansal and Yaron, 2004**).

the savings rate shrinks from 0.4 to 0.3 p.p. of GDP (0.5 to 0.3 p.p. by 2040). Informal savings fall by 20% throughout this range. Since **Ruffo** find that the amnesty reduced savings, we regard $\rho \in [1, 2]$ as the more plausible range.

6 Social Security in Rich OECD Countries

In order to study the determinants of pensions, we now calibrate the model to three rich OECD economies that differ markedly in pension spending, demography, income inequality, voting patterns and pension design: the U.S., France and the UK. In these economies nearly every worker qualifies for a contributive pension, so we set informality aside and the design reduces to one dimension, the extent to which benefits are earnings-related.¹⁷ To explore what characteristics shape pensions, savings rates, and labor supply, we simulate counterfactual economies that are calibrated to the U.S. with one characteristic changed at a time.

As for the Argentina case, the following outlines the main results. Appendix [E](#) and the [online appendix](#) contain more details on the calibration and supplementary evidence.¹⁸

Calibration. We calibrate the three economies to 2020 in the same way as Argentina in section 5, with a few exceptions. First, with informality set aside, the population is split into three formal income groups. The first two groups are cut, as for Argentina, to have mean incomes of half the mean and the mean, the incomes at which the OECD’s replacement rates identify θ . France’s system is fully Bismarckian, $\theta = 1$ at any cuts, so we group French households at the U.S. income percentiles instead, which puts the two countries’ household types in one-to-one correspondence for the counterfactuals below. Second, the discount factor β matches the U.S. thirty-year interest rate rather than a capital–output ratio and is imposed on France and the UK. Third, where the Argentine calibration gives every household the same propensity to vote, the political objective here weights each income group by its observed propensity. In the U.S. the poor vote much less than the rich, while in France the propensity to vote is nearly flat across income groups. Fourth, since the model is one of intergenerational transfers, we add Medicare to U.S. pensions, in spending and in the calibration of θ , where as a benefit unrelated to income it makes transfers to the old more Beveridgean than pensions alone.

Table 3 reports the parameters. Population growth is highest in the U.S. and lowest in France, $\nu_{2020} = 1.34$ against 0.97, and the ratio of high to low productivity is about a third higher in the U.S. than in France, 3.08 against 2.29. The average taste for leisure

¹⁷**WhitmanRa1** estimate that about 4% of individuals aged 62–84 in the U.S. in 2010 would never collect Social Security, most of them for lack of the 40 quarters of covered earnings needed to qualify.

¹⁸Appendix [E](#) holds the surveys, the shared parameters and each country’s heterogeneity by income group. The online appendix holds every table behind the figures of this section, the same counterfactuals with the UK as the host economy, and the calibrations under the alternative treatment of the taste for leisure and across the intertemporal elasticity of substitution.

is roughly twice the U.S. level in the UK and France. The calibrated political weight of the old is nearly the same in the U.S. and France, 1.45 and 1.42, and lower in the UK, 1.16.

Table 3: Calibration, the U.S., the UK and France

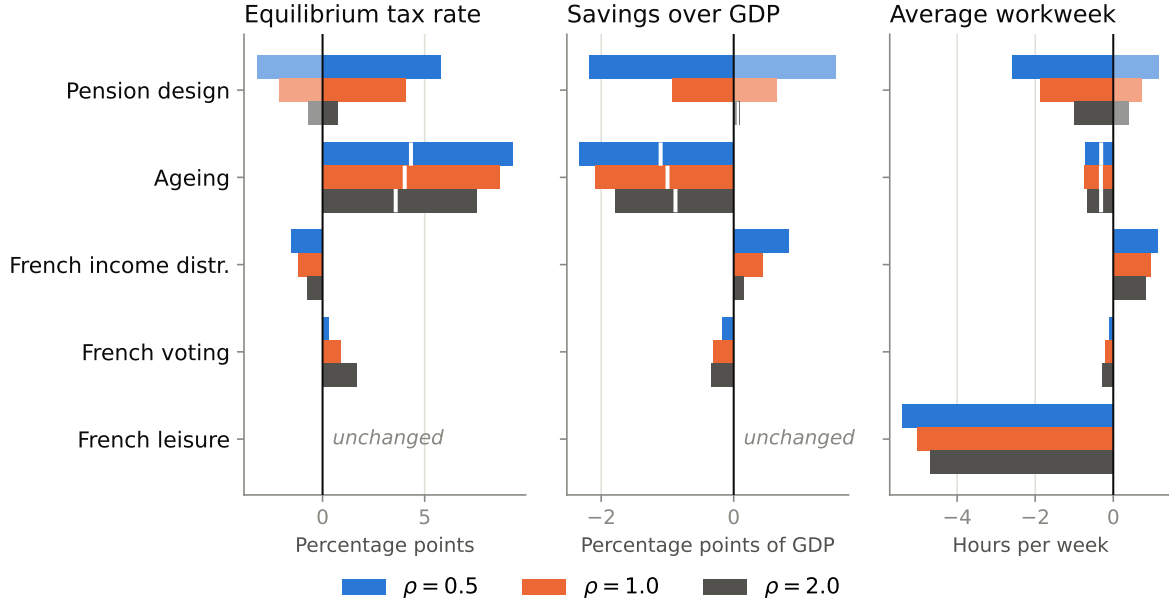
Parameter	U.S.	UK	France	Identified by
θ	0.74	0.56	1.00	Replacement rate dispersion
ω	1.45	1.16	1.42	$\tau^{US} = 14.4\%$, $\tau^{UK} = 11.9\%$, $\tau^{FR} = 21.3\%$
X	4.03	8.26	7.27	Average workweek
ν_{2020}	1.34	1.17	0.97	30-year gross population growth
η_H/η_L	3.08	3.06	2.29	Relative productivity, high to low income
β		0.76		30-year interest rate (U.S.)
ρ		1		Log-GHH preferences
ξ		0.30		Frisch elasticity of labor supply
α		0.30		Capital income share

Note: X is the population-weighted mean of X_i ; its level is the hours unit, pinned for France and the UK by targeting average hours relative to the U.S. rather than in levels. The lower panel is common to the three countries; β is calibrated for the U.S. and imposed on the other two.

Determinants of pensions, savings, and labor supply. Figure 6.1 summarizes the counterfactuals. Each row changes one characteristic of the U.S. calibration, its pension design, population growth, income distribution, leisure preferences or voting patterns, and each panel reports the effect on the equilibrium tax rate, the savings rate and the workweek in 2020, at the three values of the intertemporal elasticity ρ considered below. The counterfactuals are guided by France and the UK, particularly France, whose households can be grouped as in the U.S. and whose calibrated ω is close to the U.S. value. Each is a separate equilibrium path on which the changed characteristic holds throughout, so a row describes an economy that has always had that characteristic rather than the U.S. surprised in 2020.

The tax panel identifies the pension design and ageing as the main determinants of pension spending. Going from a fully Bismarckian system to a fully Beveridgean one raises the equilibrium tax rate by 6.2 p.p. at $\rho = 1$ and lowers the savings rate by 1.6 p.p. of GDP. The reason for this is that of proposition 2: flatter benefits direct more of the revenue to poorer retirees, whose higher marginal utility gives them more weight in the political objective, so retirees support a larger system. Holding taxes fixed, an increase in θ lowers savings, but the endogenous fall in taxes more than offsets this and the net

Figure 6.1: The effect of various counterfactuals in the U.S.



Note: Every panel is the deviation from the calibrated U.S. baseline at the same ρ , whose levels are printed in the [online appendix](#). In the pension-design row each bar shows both corners, measured from the baseline: $\theta = 0$ in full colour and $\theta = 1$ in the light tone. In the ageing row the bar is acute ageing and the white line across it mild ageing. The savings rate is savings relative to GDP.

effect is positive. For labor supply, the economic-equilibrium and political effects of an increase in θ are aligned, and the tax response contributes as much as the direct effect or more.

Population ageing is the determinant that previous studies of the politico-economic equilibrium of pension spending have highlighted: **mkvss08** predict an increase in U.S. spending of around 3.6 p.p. towards the middle of this century because of the slowdown in population growth. In the data, including projections, ageing is more acute in France than in the UK, and much more than in the U.S. We consider two scenarios, which roughly correspond to the data for the two European countries. In the first, *mild ageing*, we halve current and projected U.S. population growth; in the second, *acute ageing*, the population is constant, $\nu_t = 1$ for all t . Acute ageing raises the tax rate by 8.7 p.p. and mild ageing by 4.0 p.p.; the mild scenario is close to the demographic projection of **mkvss08**.¹⁹ With log-GHH preferences ageing moves savings only through the tax rate, which under acute ageing lowers the savings rate by 2.1 p.p. of GDP. Hours fall by at most 0.7 hours a week: in the economic equilibrium ageing deepens capital, which raises wages and labor supply, and the higher tax more than undoes this.

The characteristics that set France's households apart move the tax rate much less. France's flatter income distribution, imposed with θ held at its U.S. value, lowers it by

¹⁹In **mkvss08**, $\nu_{2000} = 1.38$ and the projected $\nu_{2080} = 1.23$, a fall of about 0.15, similar to the fall of 0.17 in 2020 in the mild ageing scenario.

1.2 p.p.,²⁰ and its flatter voting profile, under which the poor vote about as often as the rich, raises it by 0.9 p.p. Rescaling the taste for leisure to France’s level shortens the workweek by more than the whole gap to France’s and leaves every rate at the baseline, since a proportional change in X_i scales hours, output and savings together. Imposed together, the three characteristics bring the U.S. close to France’s workweek but move its tax rate slightly away from France’s, so the gap in taxes is due to France’s demography, which more than offsets its fully Bismarckian design. With the UK as the host economy the effects are smaller still, and France’s income distribution raises the UK’s tax rate, since at the UK’s income cuts France is the more unequal of the two countries (appendix E.4).

The workweek panel ranks the determinants differently. France’s leisure preferences move hours most and leave every rate unchanged. Among the characteristics that move the tax rate, the pension design moves hours most, by about two hours a week at the Beveridgean corner at $\rho = 1$, while ageing, the largest effect on taxes, moves them less than France’s income distribution does at every ρ . The determinants of pension spending are thus not the determinants of hours worked.

The effect of the intertemporal elasticity of substitution. Under general CRRA preferences, with ρ read as the IES as in section 5 and each ρ calibrated separately, the effect of ageing changes little and log preferences remain the benchmark. The exception is the pension design, whose effect on the tax rate falls steeply as the IES rises: going from $\theta = 1$ to $\theta = 0$ raises the tax rate by 9.0 p.p. at $\rho = 0.5$ and by 1.5 p.p. at $\rho = 2$. The reason for this is that a higher IES lowers the young’s resistance to taxation, so the calibration needs a lower political weight of the old to match the observed tax rate, and the difference in benefits between rich and poor retirees, through which θ moves the tax rate, counts for less; the savings rate and the workweek follow. France’s voting profile is the one characteristic whose effect grows with the IES: it raises the weight of the poor young and the poor old alike, and at a higher IES the old’s demand for a larger system meets less resistance from the young. The ranking of the determinants in figure 6.1 therefore holds at elasticities of one or below; at $\rho = 2$ France’s voting patterns move the tax rate by more than the whole range of the pension design.

The calibrated parameters absorb institutions that the model does not contain. The political weight of the old, ω , stands for the way legislation is introduced, debated and passed, and for constitutional arrangements such as whether the political system is presidential or parliamentary. The taste for leisure, X , stands for preferences and also for institutions that shape labor supply, such as unionization, collective bargaining, labor-market regulation and statutory holidays, so the large differences in X across the three

²⁰We read θ off the ratio of replacement rates at mean and half-mean income, so under France’s flatter income distribution the same observed ratio would imply a considerably less Bismarckian system. Re-estimating θ would bundle a change in the design into a counterfactual about inequality.

countries include these institutional differences.

We now return to figure 1.1, in which pension spending and the Bismarckian index both rise as population growth falls, and neither is related to income inequality. With the design given, the model accounts for the pattern in pension spending: ageing is the main determinant of the equilibrium tax rate, and the income distribution moves it several times less.²¹ The relation between the design and population growth suggests that countries respond to ageing by tying benefits more closely to earnings, which a model that holds θ fixed cannot address. Section 7 takes it up by letting the electorate choose θ as well as τ : costless redistribution puts the choice at a corner (section 7.1), and a deadweight cost of redistribution, calibrated on the U.S. design, is tested on the UK and France.

7 Endogenous Pension Design

Throughout the paper so far the Bismarckian index θ is calibrated and held fixed while the tax rate is chosen politically. This section lets the electorate choose the design as well. The design is set by the same probabilistic-voting objective $\mathcal{W}_t$ that sets τ_t , in the model without informal households, so that one political process delivers both the size of the system and its shape. We find that ageing makes the chosen design more Bismarckian and that inequality moves it little, which is the pattern of figure 1.1. The section runs in four parts: the corner that costless redistribution produces (section 7.1); the deadweight cost of redistribution that makes an interior design possible (section 7.2); its calibration on the U.S. design and the test of that calibration on the UK and France (section 7.3); and the response of the chosen design to ageing, inequality and voting patterns (section 7.4). Derivations, the algorithms and the verification battery are in the technical documentation published alongside the paper.²²

7.1 Costless redistribution: a corner

We consider three timings of the choice, and each puts the design at a corner. If θ_t is chosen at t together with τ_t , it re-splits benefits that current retirees have already earned. Current labor supply depends on the design that will apply to the benefits it earns, θ_{t+1} , so the choice moves no economic decision, and its marginal effect on the political objective

²¹On the disposable-income Gini of figure 1.1, the U.S., at 0.42, is the third most unequal of the 29 countries, after Mexico and Turkey, and France, at 0.32, is the median.

²²**BergGE26doc**, section *Endogenous system characteristics*: the model definitions and propositions for each timing of the choice, the derivation of the cost, the numerical algorithms, including the exact two-dimensional policy-function recursion under CRRA preferences, and the calibration protocol.

runs entirely through current retirees,

$$\frac{\partial \mathcal{W}_t}{\partial \theta_t} = \omega \sum_i \gamma_i \mu_i \frac{\frac{1-\alpha}{\alpha} \tau_t (y_i - 1)}{\frac{s_{t-1}^i}{s_{t-1}} + \frac{1-\alpha}{\alpha} \tau_t (1 + \theta_t [y_i - 1])}, \quad y_i \equiv \frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_h}, \quad (15)$$

where y_i is the labor income of type i relative to the mean. The numerator is the change in the benefit of type i and averages to zero across types: a more Bismarckian split moves a given total from retirees below mean income to retirees above it. The denominator is proportional to the consumption of a retiree of type i and rises with income. With equal propensities to vote the sum therefore weights the gains of the rich by a lower marginal utility than the losses of the poor, and is negative at every $\theta_t \in [0, 1]$ whatever the tax rate, the political weight of the old and the demography: the electorate chooses the Beveridgean corner. We prove this in proposition 3 (appendix B.4) for any propensities to vote that do not rise with income, equal ones included: the derivative is negative at every design and the design chosen with the tax is $\theta_t = 0$, whatever the tax rate, the political weight of the old, the demography, the income distribution and the intertemporal elasticity, and whatever design current retirees expected when they saved. Only propensities to vote that rise with income can overturn the corner. They do in the U.S. calibration of section 6, so there the sign is a quantitative matter: evaluated at the calibrated 2020 equilibrium the expression is negative over the whole interval, at every date of the projection and at $\rho = 0.5, 1$ and 2 , and the choice is the corner $\theta = 0$.

If instead θ_{t+1} is chosen at t , the design no longer touches the benefits of current retirees but moves current labor supply and savings, directly and through the anticipated response of τ_{t+1} . With the political objective of (14) the marginal effect is

$$\frac{\partial \mathcal{W}_t}{\partial \theta_{t+1}} = \sum_i \gamma_i \mu_i \left[\omega (1 - \alpha) \frac{d \ln \Theta_{h,t}}{d \theta_{t+1}} + \nu_t \left((1 + \beta) \frac{d \ln \tilde{c}_{1,t}^i}{d \theta_{t+1}} + \beta \frac{d \ln R_{t+1}}{d \theta_{t+1}} \right) \right], \quad (16)$$

where $\Theta_{h,t}$ is the labor supply function of appendix A.2 and each total derivative includes the response of τ_{t+1} , and of θ_{t+2} , to the design chosen. Retirees are unanimous, since their term does not depend on i , and they favour a more contributive system: a higher θ_{t+1} raises the return to an hour worked at t and with it the labor supply on which their benefits are levied. The young are divided between this forward-looking stake and the redistribution of their own future benefits, the force of (15) one period ahead. The equilibrium is a sequence of policy functions over the inherited design, identified by backward induction as in section 4, and at observed U.S. inequality the redistributive force dominates: the choice is again the corner $\theta = 0$.

A design chosen once and for all enters every channel at once, and the outcome is again a corner: $\theta = 0$ under log preferences, and under CRRA preferences the Bismarckian corner $\theta = 1$ at high intertemporal elasticities, where the forward-looking channels are

strong.²³ The common lesson is that a costless choice never lands in the interior: some friction must make flat benefits more expensive than earnings-related ones for an interior design to survive.

7.2 A deadweight cost of redistribution

We introduce a deadweight cost of the redistribution that the flat component of benefits performs, so that the government budget becomes

$$\begin{aligned} \nu_t w_t \tau_t h_t f(\theta_t, \tau_t) &= \sum_i \gamma_i b_t^i, \\ f(\theta_t, \tau_t) &= \exp\left(-\frac{1}{2} \lambda \tau_t \tilde{V} (1 - \theta_t)^2\right), \quad \tilde{V} \equiv \sum_i \gamma_i \frac{(y_i - 1)^2}{y_i}, \end{aligned} \tag{17}$$

where $\lambda \geq 0$ is the cost parameter and a share $1 - f$ of revenue does not reach retirees. Nothing is lost when nothing is redistributed: $f = 1$ at $\theta_t = 1$, and $f = 1$ for every design when all incomes are equal, where a flat and an earnings-related benefit pay the same to everyone. The parameter λ is a reduced form for whatever makes redistribution through the pension system expensive, and its discipline is the cross-country test of section 7.3: one value, calibrated on the U.S. design alone, has to place the designs of the UK and France.

The form follows from reading the flat component as a tax. Part of a pension contribution buys the contributor a benefit and part does not, and only the second part distorts like a tax, the tax component of contributions of **Summers89a1**. Relative to a purely earnings-related system paying the same revenue, the flat component moves the benefit of type i by $\tau_t(1 - \theta_t)(1/y_i - 1)$ per unit of past earnings: a tax on types above mean income, a subsidy to types below it, and a pure transfer in the aggregate.²⁴ The model contains the hours response to this wedge, through the Frisch elasticity; the cost stands for the margins it does not contain, on which the evidence finds the tax component to matter: participation, where **Disney04a1** finds that the tax component of contributions in OECD countries lowers the participation of women while the saving component raises it, and reporting, where **KumlerVF20a1** find that Mexico's 1997 reform, which tied benefits more closely to reported wages, reduced the underreporting of wages. On both margins the wedge that matters is the average tax on the earnings at stake rather than the marginal one (**Saez02a1**; **KlevenK06a1**), and the average implicit tax of the flat

²³Near the elasticity at which the corner switches the electorate is close to indifferent between the two, so the switch is not sharply located. The joint choice of τ_{t_0} and θ reduces to a one-dimensional search, and the model with a permanent choice can be calibrated to the observed 2020 characteristics only under counterfactually compressed inequality; the technical documentation gives the algorithm and its checks.

²⁴Up to the pay-as-you-go return factor $\Pi_t = \nu_t w_t h_t / (w_{t-1} h_{t-1})$, which multiplies every implicit tax alike and which we absorb into λ , so that an elasticity ε corresponds to $\lambda = \Pi_t \varepsilon$. The factor is of the order of ν_t , which is 1.34 for the U.S. in 2020 (table 3). The technical documentation gives the derivation.

component differs across types.

The deadweight loss of a set of such wedges is the sum of their **Harberger64a1** triangles. With a common elasticity ε it is $\frac{1}{2}\varepsilon\tau_t(1-\theta_t)^2\tilde{V}$ as a share of revenue, which is $1-f$ in (17) to first order with λ in the place of ε .²⁵ The loss is quadratic in the flat share $1-\theta_t$ because it is quadratic in each wedge; proportional to the dispersion of relative incomes $\tilde{V}$ because the wedges scale with the distance of incomes from the mean, so a compressed income distribution lowers the redistribution a flat benefit performs and the cost of performing it together; and linear in the size of the system τ_t per unit of revenue, since the loss rises with the square of the implicit tax while revenue rises with the tax itself. The last property makes a larger system costlier to keep redistributive, the efficiency–redistribution trade-off by which **KoethenbuergerPP08a1** account for the finding of **CondeRuizP07a1** that more redistributive systems are smaller, read from the other side. The shape of f follows from the tax component, but its size does not: at $\rho \leq 1$ the calibrated λ (table 4) is more than an order of magnitude above what participation and taxable-income elasticities of 0.1 to 0.4 would give (**ChettyGMW11a1**; **SaezSG12a1**), and at $\rho = 2$ roughly three to thirteen times as large, so λ also stands in for administrative, evasion and political costs of redistribution that the model does not contain.²⁶

With the cost, the benefit formula carries $A(\theta_t, \tau_t) = f\theta_t$ and $B(\theta_t, \tau_t) = f(1-\theta_t)$ in place of θ_t and $1-\theta_t$, and since the design enters the economic equilibrium only through these two groups, the cost is exactly that substitution in the equilibrium functions of appendix A.2. Because the cost is proportional it cancels from the replacement-rate ratio that identifies θ from the OECD data, so the data target is unchanged, $\theta^* = 0.738$. In the tax condition the cost adds one factor: since f falls with the tax that funds it, the benefit term of the retirees’ marginal effect $\mathcal{E}_{2,t}^i$ in appendix B.1 is multiplied by $1 + \tau_t \partial_\tau \ln f = 1 + \ln f$, so a larger system is worth less to a retiree at the margin than the revenue it collects.

We let the design be chosen one period in advance, and use only this timing with the cost. A design chosen for the current period moves no labor supply or savings decision, so the cost would be the only force against the corner of (15); a design chosen once and for all requires a commitment that the Markov equilibria of the rest of the paper do not have. The choice one period in advance keeps the Markov structure, it is the only timing in which the design affects labor supply and savings before it binds, the channel

²⁵The quadratic counts the subsidy to types below mean income as a loss, as it does the tax on types above it. Where participation is already taxed, lowering the participation tax of low-income workers can reduce the total loss (**Saez02a1**), so the symmetry is a property of the reduced form rather than of the argument.

²⁶The cost is proportional, scaling both components of the benefit alike. A cost on the flat component only delivers the same endpoints and differs in the interior, at a cost in identification, since f then no longer cancels from the replacement-rate ratio and θ^* and λ are identified jointly. A cost attached to the design itself rather than to the transfer it performs, $f(\theta) = \phi + (1-\phi)\theta^\rho$, stays in place when a compressed income distribution removes the redistributive stakes; the [online appendix](#) reports it as a robustness check.

through which θ moves taxes and savings in section 6, and it mirrors how reforms of benefit formulas are legislated, applying to future benefits and leaving benefits in payment untouched.

In this timing the design in force, θ_t , is a state whose cost enters the benefits of current retirees and, through them, the tax condition; the design chosen at t , θ_{t+1} , enters (16) through next period's groups $A(\theta_{t+1}, \tau_{t+1})$ and $B(\theta_{t+1}, \tau_{t+1})$, with τ_{t+1} the anticipated equilibrium tax. What makes an interior choice possible is how the two groups move with the design. Without the cost a more Bismarckian design shifts next period's benefits one for one from the flat to the earnings-related component, $\partial_\theta A = -\partial_\theta B = 1$, and the redistributive stake of the young dominates, as in section 7.1. With the cost, $\partial_\theta A + \partial_\theta B = f_\theta \geq 0$, where $f_\theta = f \lambda \tau_{t+1} \tilde{V}(1 - \theta_{t+1})$ at a given tax: a more Bismarckian design also raises the total the young will receive, since it dissipates less. This gain is zero at $\theta_{t+1} = 1$ and rises with the flat share, the dispersion of incomes and the size of the system the young anticipate, so the design condition can cross zero in the interior. The explicit expressions, and the proofs that the substitution and the state independence of the log case survive the cost, are in the technical documentation.

7.3 Calibration and a cross-country test

The cost parameter is identified separately from the design, by requiring that the design in force in the calibration year, on a path where the political choice binds from the first period, is the observed one,

$$\theta_{t_0} = \Theta_{t_0-1}(\theta^*) = \theta^*, \quad (18)$$

with (β, ω) recalibrated at every trial value of λ . The calibration pins the design in the calibration year only; every later design is a prediction of the model. Table 4 reports the calibrated cost across the paper's range of intertemporal elasticities, at the U.S. dispersion of relative incomes $\tilde{V} = 0.436$: $\lambda = 18.267, 8.643$ and 1.724 at $\rho = 0.5, 1, 2$, a fall by a factor of about ten. At those values the observed design loses 3.9%, 1.8% and 0.4% of revenue at the 2020 size of the system, and a fully flat system of the same size would lose 44%, 24% and 5%. The reason for this is that a higher elasticity strengthens the forward-looking channels of (16), so less friction is needed to hold the choice away from the Beveridgean corner. That the calibrated cost lies closest to estimated elasticities at $\rho = 2$ adds to the case of section 5 for $\rho \in [1, 2]$ as the more plausible range.

The cost is calibrated on one datum, the U.S. design, so the UK and France are a test of the specification rather than of its fit. Table 5 lets each electorate choose the design on its own calibrated path under the U.S. parameter $\lambda = 8.643$, at $\rho = 1$. The UK, with an observed design of 0.560 and a tax rate of 11.9%, chooses 0.625; France, with the corner $\theta = 1$ and 21.3%, chooses 0.749; the U.S. re-elects 0.738 by construction. The observed

Table 4: The calibrated cost of redistribution

CRRA (ρ)	λ	θ^*	$f(\theta^*)$	$f(0)$	$\tilde{V}$
0.5	18.267	0.738	0.961	0.563	0.436
1.0	8.643	0.738	0.982	0.762	0.436
2.0	1.724	0.738	0.996	0.947	0.436

Note: $f(\theta, \tau) = \exp\{-\frac{1}{2}\lambda\tau\tilde{V}(1-\theta)^2\}$ with $\tilde{V} = \sum_i \gamma_i(y_i - 1)^2/y_i$ the dispersion of relative labor incomes; $f(\theta^*)$ and $f(0)$ are read at the 2020 tax rate, so $1 - f$ is the share of revenue lost at the observed design and at a flat benefit; λ is calibrated so the design in force in 2020 is the observed one, with (β, ω) recalibrated at each trial value. The cost is proportional, so it cancels from the replacement-rate ratio and θ^* is the data's own. Without the cost the choice would be in the corner $\theta = 0$ for every ρ .

ranking of the three is thus reproduced from one parameter, though compressed. The reason for this is the size of the system: the UK at its own income cuts and France at the U.S. percentiles have nearly the same dispersion of relative incomes, $\tilde{V} = 0.223$ and 0.218, about half the U.S. value, and what separates their chosen designs is the tax that funds them, which is the relation between the size of the system and its design that figure 1.1 shows. Calibrating the UK's own cost, so that its electorate re-elects 0.560, gives $\lambda = 7.264$ at $\rho = 1$, within 16% of the U.S. value, and 17% below it at $\rho = 0.5$; the agreement breaks at $\rho = 2$, where the UK's own cost, which falls less steeply in the elasticity than the U.S. one, lies well above it (online appendix). Regrouping the UK's households at the U.S. income percentiles, which lowers its $\tilde{V}$ to 0.065, moves the choice under the U.S. parameter only to 0.644, so the result does not turn on how the income groups are cut. France has no own cost: its observed design is the corner, and under a cost quadratic in the redistribution performed the first unit of redistribution is free at the margin, so no electorate chooses exactly $\theta = 1$.²⁷

7.4 Ageing, inequality and voting patterns

By construction the design in force in 2020 is θ^* at every ρ . Along the projected demographic path the chosen design then drifts toward earnings-related benefits as ageing proceeds, tracking the projected fall in ν_t and flattening when it does, while the tax rate rises a little above the path with the design held at θ^* , since a more Bismarckian system dissipates less and is worth more to the electorate at the margin.²⁸

²⁷A linear term in $1 - \theta$ would restore that possibility at the price of a second parameter, which the UK's design could pin with France as the prediction; we leave it for future work.

²⁸The design channel runs through hours, so the Frisch elasticity is the parameter to vary. Recalibrating (β, ω) and the cost at $\xi = 0.2$ and 0.4 moves λ by about one percent from its value at $\xi = 0.3$, to 8.736 and 8.547, and leaves the design in force in 2110 unchanged to the third decimal and the design chosen under acute ageing within one unit of it; only the level of the tax moves, by about 0.3 p.p. in either direction by 2110.

Table 5: The UK and France under the U.S. cost of redistribution

Economy	θ^* observed	Tax rate	θ chosen, U.S. λ	Own λ	$\tilde{V}$
UK	0.560	11.9%	0.625	7.264	0.223
France	1.000	21.3%	0.749	—	0.218
UK at U.S. income groups	0.543	11.9%	0.644	6.519	0.065

Note: The deadweight cost $f(\theta_t, \tau_t)$ of equation (17) is quadratic in the implicit tax the flat component levies and scaled by the size of the system, $f = \exp\{-\frac{1}{2}\lambda\tau_t\tilde{V}(1 - \theta_t)^2\}$, with λ calibrated per ρ (table 4). Each economy is its own calibration at $\rho = 1$ (β imposed from the U.S., ω its own; table 3); the UK at U.S. income groups is the UK workbook regrouped at the U.S. income percentiles. “Chosen” is the design in force in 2020 on the economy’s own freely simulated path under the U.S. parameter $\lambda = 8.643$, to be read against the observed one; “own” is the value at which that path re-elects the observed design, with ω recalibrated at each trial value. No finite λ makes France re-elect its observed design: it is the corner $\theta = 1$, and under a cost quadratic in the redistribution performed the first unit of redistribution is free at the margin.

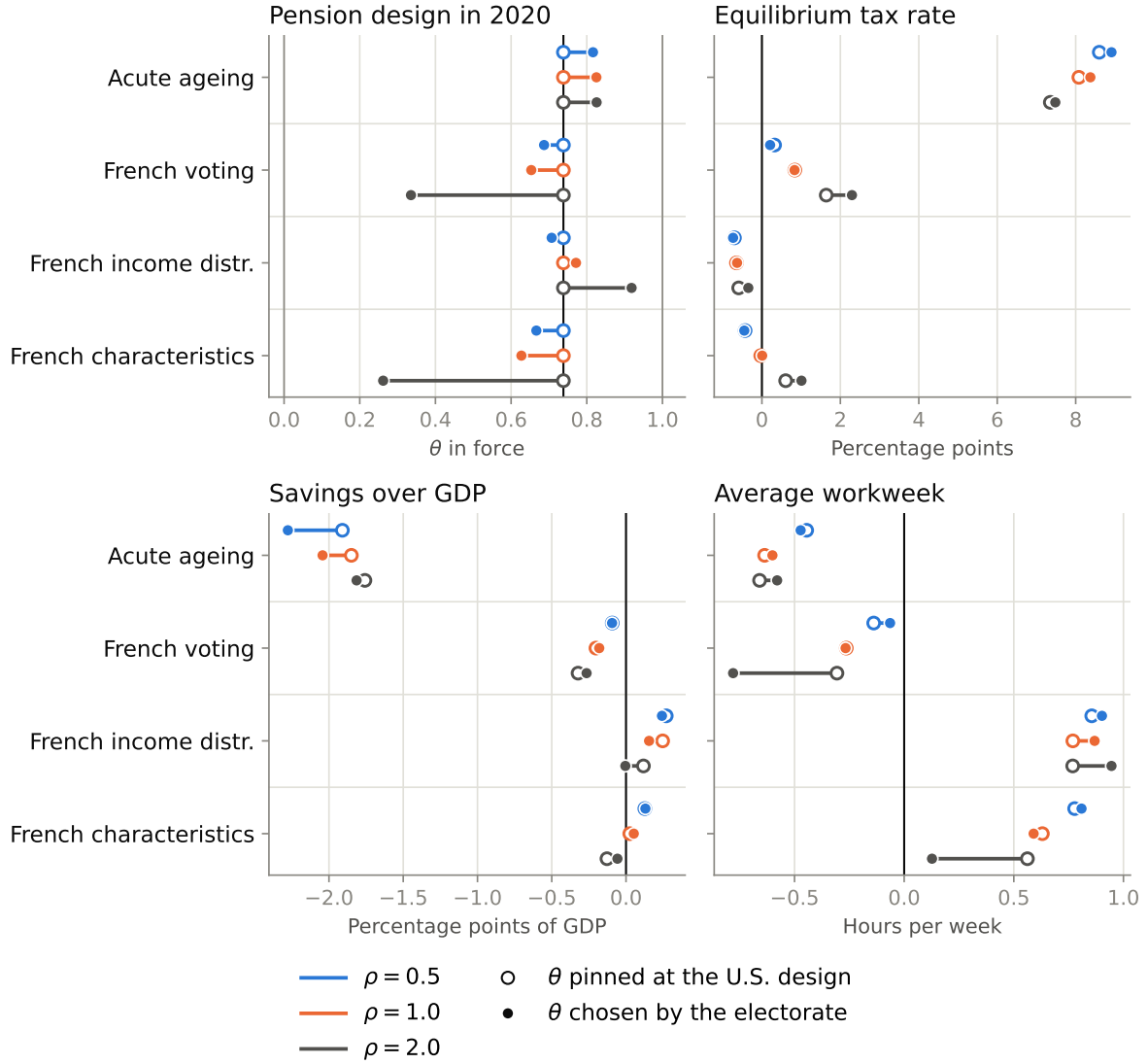
Each counterfactual below is a separate equilibrium path read at 2020, with the changed characteristic holding over the whole horizon and the political choice binding from the first period, so a row describes a country that has always had this mix of characteristics rather than the U.S. surprised in 2020. Table 10 in appendix F and the tables of the [online appendix](#) report each counterfactual with the design pinned at the U.S. value and with the design chosen, at $\rho \in \{0.5, 1, 2\}$; both readings carry the cost, so the distance between them is the political response.²⁹

Figure 7.1 collects the whole set: an open marker is the reading with the design pinned, a filled marker the reading with the design chosen, and the length of the connector is what endogenizing the design adds. At $\rho \leq 1$ the design response moves hours by at most 0.1 hours a week in every counterfactual, so the hours effect of a counterfactual is its exogenous one; at $\rho = 2$ the design moves far enough under French voting patterns to cost about half an hour a week, 38.6 against 39.1 hours with the design pinned ([online appendix](#)).

Ageing. Acute ageing, $\nu_t = 1$ throughout, raises the chosen design from 0.738 to 0.825 at $\rho = 1$, and to 0.816 and 0.826 at $\rho = 0.5$ and 2 (table 10), so the response varies little with the elasticity, unlike the calibrated cost. Two mechanisms add up: a scarcer young generation raises the political weight of retirees, who are unanimous for a more contributive system in (16), and the tax that ageing calls for makes the flat component costlier at the margin, since the loss in (17) is proportional to τ_t . Under acute ageing the

²⁹We report the chosen design under the common- X calibration of section 6 only; the design and the cost enter the political objective through relative incomes alone, which both calibrations match to the same data, so only the income-distribution rows would read differently against the vector- X_i calibration of the [online appendix](#).

Figure 7.1: Endogenous versus exogenous pension design across counterfactuals



Note: The first panel is the design in force as a level, with the muted lines at the corners $\theta = 0$ and $\theta = 1$; the other three are deviations from the endogenous- θ baseline, whose levels are the baseline rows of table 10. The savings rate is savings relative to GDP.

tax rises from 14.4% to 22.8%, so the loss per unit of redistribution rises by more than half, and this second channel is what makes the response first order. The tax and savings responses are essentially those of the exogenous- θ reading: at $\rho = 1$ the chosen design adds 0.3 p.p. to a tax response of 8.1 p.p., and between 0.1 and 0.3 p.p. across ρ , since a more Bismarckian system dissipates less and is worth a larger tax to the electorate. The pinned response is a little below the 8.7 p.p. of section 6, because with the cost in place a larger system is worth less to retirees at the margin.

Income distribution. The model's inequality is the dispersion of relative labor income across the three income groups at the chosen cuts, the $\tilde{V}$ of (17), a different object from the Gini coefficient of figure 1.1. Holding θ at the U.S. value, France's flatter income

distribution lowers the equilibrium tax at $\rho = 1$ from 14.4% to 13.8% ([online appendix](#)), less than the 1.2 p.p. of section 6, because a compressed distribution also lowers $\tilde{V}$ and with it the share of revenue the flat component dissipates. When the electorate chooses the design it moves to 0.772 at $\rho = 1$, and to 0.707 and 0.918 at $\rho = 0.5$ and 2: interior at every elasticity and rising in it. This happens because the cost is proportional to the same dispersion of relative incomes as the redistribution a flat benefit performs, so compressing the distribution lowers the redistributive stakes and the cost of carrying them together; what remains is a small net effect whose sign depends on the elasticity, since the young's forward-looking stake in earnings-related benefits does not scale with the dispersion of incomes and gains ground when the redistributive stakes shrink. The design response leaves the tax where the pinned reading puts it at $\rho \leq 1$ and adds 0.2 p.p. at $\rho = 2$, and savings move by less than 0.2 p.p. of GDP. The model therefore does not tie the design tightly to inequality, in line with the weak relation between inequality and the Bismarckian index across countries in figure 1.1.

Voting patterns. France's much flatter voting-weight profile pushes the design toward the Beveridgean corner, and this is the experiment whose strength depends most on the elasticity: the chosen design falls from 0.738 to 0.653 at $\rho = 1$, and to 0.687 and 0.335 at $\rho = 0.5$ and 2 ([online appendix](#)). When the poor carry more relative voting weight, the redistributive force of (15) regains ground against the forward-looking stake; at $\rho = 1$ the response is of the same size as that of acute ageing, with the opposite sign. That the design falls furthest at the highest elasticity follows from the calibration: the cost that re-elects the observed design amounts to only 0.4% of revenue at $\rho = 2$ against 3.9% at $\rho = 0.5$ (table 4), so at a high elasticity the interior choice rests on a thin cushion and a stronger redistributive pull moves it a long way. In the exogenous reading a flatter voting profile makes the system larger, by 0.8 p.p. at $\rho = 1$; the design response leaves that tax within 0.1 p.p. at $\rho \leq 1$, because the less Bismarckian design chosen also dissipates more revenue, and lifts it to 16.7% against 16.1% at $\rho = 2$, where the design moves a long way and the cost is thin.

French characteristics. When all of France's household characteristics are imposed together, the voting profile prevails at every elasticity: the chosen design is 0.627 at $\rho = 1$, and 0.667 and 0.262 at $\rho = 0.5$ and 2, with the tax at 14.4% in both readings at $\rho = 1$ ([online appendix](#)); France's level of X_i contributes nothing to the design or the tax, since a common rescaling of X_i moves hours and nothing else (section 6). What the characteristics do not deliver is France's own corner, for the reason section 7.3 gives. The same exercise with the UK as the host economy, at the UK's own cost parameter, moves its design toward flat benefits under France's voting profile as well, by less, and leaves its tax rate within 0.1 p.p. of the pinned reading in every row ([online appendix](#)), so on the UK the design response is confined to the design itself. The two forces that figure 1.1

motivates thus come out as the data do: ageing moves the chosen design a long way, and inequality moves it little.

8 Conclusions

This paper studies how two features of a pension system, whether benefits require a contribution record and how closely they follow past earnings, shape its political support and through it taxes, savings and labor supply. The setting is an overlapping generations model with capital, heterogeneous households, an informal sector and probabilistic voting, first over the size of the system and then over its design as well.

With log-GHH preferences and hand-to-mouth informal households the equilibrium tax is a function of parameters and demographics alone. It rises with the political weight of the old, with ageing and with income inequality, and it falls when benefits are more closely tied to earnings, since earnings-related benefits move revenue from poor retirees to rich ones; universal pensions raise the tax when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption. Calibrated to Argentina, the model accounts for about three quarters of the rise in pension spending that followed the expansion of coverage of 2005 to 2010, with the savings rate lower by 0.4 p.p. of GDP and the workweek by a quarter of an hour. Calibrated to the U.S., France and the UK, ageing is the main determinant of the tax rate and the earnings link the second; income inequality and voting patterns matter little, and a higher intertemporal elasticity of substitution changes only the size of the earnings-link effect.

When the electorate chooses the earnings link as well, costless redistribution puts the choice at a corner: with propensities to vote that do not rise with income the design chosen with the tax is fully Beveridgean, and at U.S. inequality every timing of the choice gives a corner. An interior design needs flat benefits to be costlier than earnings-related ones. With a deadweight cost of redistribution that rises with the size of the system, calibrated on the U.S. design alone, ageing makes the chosen design more Bismarckian, inequality moves it little, and the single cost parameter orders the designs of the UK, the U.S. and France as observed; the UK's own cost is within 17% of the U.S. value at intertemporal elasticities of one or below, though not at two. This is the pattern of the OECD cross-section: spending and the earnings link rise together as population growth falls, and neither follows inequality.

Two limitations point to future work. The model has no risk; in **BergGEwp** we study a related environment in which health risk may prevent households from qualifying for a contributive pension, and find that households work and save more in response. And the cost specification cannot deliver France's fully Bismarckian system, because under a cost quadratic in the redistribution performed the first unit of redistribution is free at the margin; a linear term in the cost, or a formality margin, toward which the informal

sector of our Argentine model is a first step, would allow it. The same margin bears on the rise of platform work and other arrangements outside contributive coverage in rich economies, which the model can address.

Appendices

A Economic Equilibrium

The appendix derives results for the economic equilibrium with CRRA-GHH and log-GHH preferences and where informal workers' endowment is tied to wages in the formal sector absent taxes.

A.1 With CRRA-GHH preferences

Household solutions With CRRA-GHH preferences, the first order conditions in (6) can be written as

$$h_{t,i} = \left(\frac{\eta_i w_t (1 - \tau_t) + \frac{\theta \bar{b}_{t+1} \eta_i}{R_{t+1}}}{X_i} \right)^\xi$$

$$c_{2,t+1}^i = \tilde{c}_{1,t}^i (\beta R_{t+1})^\rho.$$

Using the budget constraints and rearranging slightly, we can write individual savings as

$$s_{t,i} = \frac{B_{t+1}}{1 + B_{t+1}} \left[w_t (1 - \tau_t) h_{t,i} \eta_i - \frac{X_i \xi}{1 + \xi} (h_{t,i})^{\frac{1+\xi}{\xi}} \right] - \frac{b_{t+1}^i / R_{t+1}}{1 + B_{t+1}},$$

where we have used the shorthand notation $B_{t+1} \equiv (\beta)^\rho (R_{t+1})^{\rho-1}$.

When informal households save using informal technology, similar conditions apply:³⁰

$$h_{t,0} = \left(\frac{\eta_0 w_t^0}{X_0} \right)^\xi \tag{A.1}$$

$$c_{2,t+1}^0 = \tilde{c}_{1,t}^0 (\beta R_{t+1}^0)^\rho. \tag{A.2}$$

As for formal household we can derive the following for informal savings:

$$s_{t,0} = \frac{B_{t+1}^0}{1 + B_{t+1}^0} \left[w_t^0 h_{t,0} \eta_0 - \frac{X_0 \xi}{1 + \xi} (h_{t,0})^{\frac{1+\xi}{\xi}} \right] - \frac{b_{t+1}^0 / R_{t+1}^0}{1 + B_{t+1}^0},$$

where we have used $B_{t+1}^0 \equiv (\beta)^\rho (R_{t+1}^0)^{\rho-1}$. When informal households are hand-to-mouth and receive an endowment when old, the solution for labor supply is given by (A.1).

Savings and labor supply in equilibrium. Next, in economic equilibrium, we impose conditions for factor prices in equations (3) and the specification of pension benefits

³⁰Note that given our assumption on universal pensions, informal workers do not perceive a link between their work effort and the benefits they might receive.

including the balanced budget condition in equations (8). We can then rewrite type i 's labor supply decision as

$$h_{t,i} = \left(\frac{\eta_i}{X_i}\right)^\xi \left(\frac{1}{h_t}\right)^\xi \left((1-\alpha)(1-\tau_t) \left(\frac{s_{t-1}}{\nu_t}\right)^\alpha h_t^{1-\alpha} + \frac{1-\alpha}{\alpha} \frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \right)^\xi.$$

The only type-specific term is $(\eta_i/X_i)^\xi$; this shows that the ratio $h_{t,i}/h_t$ is only a function of parameters, namely that

$$\frac{h_{t,i}}{h_t} = \frac{(\eta_i/X_i)^\xi}{\Gamma_h}, \quad \Gamma_h \equiv \sum_{i>0} \frac{\gamma_i \eta_i^{1+\xi}}{X_i^\xi}. \quad (\text{A.3})$$

Next, we can use the individual labor decision and wage rates to write

$$(1-\alpha)(1-\tau_t)h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t}\right)^\alpha = \frac{h_t^{\frac{1+\xi}{\xi}}}{\Gamma_h^{1/\xi}} - \frac{1-\alpha}{\alpha} \frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1},$$

Using this and the discounted pension benefits in equilibrium, we have that

$$s_{t,i} = \frac{1}{1+B_{t+1}} \left[B_{t+1} \frac{\eta_i^{1+\xi}}{X_i^\xi} \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \frac{1}{1+\xi} - \frac{1-\alpha}{\alpha} \frac{1-\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \right] - \frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_h} \frac{1-\alpha}{\alpha} \frac{\theta}{1+\gamma_0\epsilon} s_t \tau_{t+1} \quad (\text{A.4})$$

Using the definition of the aggregate state (1) and rearranging, we then have:

$$s_t = \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}, \quad \Gamma_{s,t} \equiv \frac{1}{1+\xi} \frac{\Gamma_h B_{t+1}}{1+B_{t+1} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} (1+\theta B_{t+1})} \quad (\text{A.5})$$

Furthermore, note that we can combine equations for s_t, h_t to yield a closed form expressions *given* values of B_{t+1} . From the definition of h_t we use that s_t is proportional to $h_t^{(1+\xi)/\xi}$ to derive:

$$h_t = \Theta_{h,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha\xi}{1+\alpha\xi}}, \quad \Theta_{h,t} \equiv (\Gamma_h)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{(1-\alpha)(1-\tau_t)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon} \Gamma_{s,t}} \right)^{\frac{\xi}{1+\alpha\xi}} \quad (\text{A.6})$$

Using this, an alternative expression for s_t is:

$$s_t = \Theta_{s,t} \left(\frac{s_{t-1}}{\nu_t}\right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}}, \quad \Theta_{s,t} \equiv \left(\frac{\Theta_{h,t}}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} \Gamma_{s,t}. \quad (\text{A.7})$$

Using the relation between s_t and $h_t^{\frac{1+\xi}{\xi}}$ in (A.6), the ratio of individual (A.4) to aggregate

savings satisfies:

$$\frac{s_{t,i}}{s_t} = \frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_h} \frac{1}{\Gamma_h} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} \frac{1-\theta}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_h} - 1 \right). \quad (\text{A.8})$$

For informal households, we assume that informal wages are proportional to formal wages absent taxes and scaled appropriately by informal productivity η_0 . Furthermore, we assume that the return to informal savings is proportional to the formal return on savings $R_t^0 = r_0 R_t$.

Specifically, we assume that informal wages are given by:

$$w_t^0 = \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha}{1+\alpha\xi}}$$

Informal labor supply then follows directly from the derivations above for formal workers; in the case with informal savings, the ratio of informal to aggregate savings is similarly given by

$$\frac{s_{t,0}}{s_t} = \frac{B_{t+1}^0}{1+B_{t+1}^0} \frac{\eta_0^{1+\xi}}{X_0^\xi (1+\xi)} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} \frac{1}{\Theta_{s,t}} - \frac{1}{1+B_{t+1}^0} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} \quad (\text{A.9})$$

Consumption functions. It is straightforward to derive relevant equilibrium consumption functions $(c_{1,t}^j, \tilde{c}_{1,t}^j, c_{2,t}^j)$ given the results above. Imposing household budget constraints (5), factor prices (3), pension benefits and balanced budget condition (8), and labor and savings decisions from above, we have for formal households i :

$$c_{1,t}^i = \frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_{s,t}} \left(1 - \frac{B_{t+1}}{(1+B_{t+1})(1+\xi)} \right) + \frac{s_t}{1+B_{t+1}} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \quad (\text{A.10})$$

$$c_{2,t}^i = \alpha \nu_t h_t^{1-\alpha} \left(\frac{s_{t-1}}{\nu_t} \right)^\alpha \left(\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{\tau_t}{1+\gamma_0\epsilon} \left(1 + \theta \left[\frac{\eta_i^{1+\xi}}{X_i^\xi \Gamma_h} - 1 \right] \right) \right) \quad (\text{A.11})$$

$$\tilde{c}_{1,t}^i = \frac{s_t}{1+B_{t+1}} \left(\frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{(1+\xi)\Gamma_{s,t}} + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \right) \quad (\text{A.12})$$

When informal households save using informal technology, equilibrium consumption functions are given by

$$\tilde{c}_{1,t}^0 = \frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^\xi} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{1}{1+B_{t+1}^0} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} s_t \quad (\text{A.13})$$

$$c_{2,t}^0 = r_0 \alpha \nu_t \left(\frac{s_{t-1}}{\nu_t} \right)^\alpha h_t^{1-\alpha} \left(\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_t}{1+\gamma_0\epsilon} \right). \quad (\text{A.14})$$

When informal households are hand-to-mouth, relevant equilibrium consumption func-

tions are given by³¹

$$\tilde{c}_{1,t}^0 = \frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^\xi} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} \quad (\text{A.15})$$

$$\tilde{c}_{2,t}^0 = \frac{1}{1+\xi} \frac{(\chi\eta_0)^{1+\xi}}{X_0^\xi} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha(1+\xi)}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon} \left(\frac{s_{t-1}}{\nu_t} \right)^\alpha h_t^{1-\alpha}. \quad (\text{A.16})$$

A.2 With log-GHH preferences

With log-preferences, the economic equilibrium is identical to the CRRA equations outlined in appendix A.1, except for the discount function $B_t = B_t^0 = \beta$. All relevant variables $x_t \in \{s_t, h_t, s_t^i, h_t^j, \tilde{c}_{1,t}^j, c_{2,t}^i, \tilde{c}_{2,t}^0, c_{2,t+1}^i, \tilde{c}_{2,t+1}^0\}$ can be written on the form $x_t = \Theta_{x,t} \left(\frac{s_{t-1}}{\nu_t} \right)^{\sigma_x}$ where σ_x are parameter functions

$$\begin{aligned} \sigma_h &= \sigma_h^j = \frac{\alpha\xi}{1+\alpha\xi} \\ \sigma_{\tilde{c}_{1,t}}^j &= \sigma_{c_{2,t}}^i = \sigma_{\tilde{c}_{2,t}}^0 = \sigma_{s,t} = \sigma_{s,t}^i = \frac{\alpha(1+\xi)}{1+\alpha\xi} \\ \sigma_{c_{2,t+1}}^i &= \sigma_{\tilde{c}_{2,t+1}}^0 = \left(\frac{\alpha(1+\xi)}{1+\alpha\xi} \right)^2, \end{aligned}$$

and where functions $\Theta_{x,t}$ only depends on parameters and policy.

Proof of proposition 1. Part *i.* outlines the marginal effect on s_t, h_t from policy and pension characteristics. Given the solution structure, the effect on s_t, h_t can be found by differentiating equations (A.5) and (A.6) with respect to the relevant variable. For τ_t the effect is straightforward:

$$\begin{aligned} \frac{\partial \ln(h_t)}{\partial \tau_t} &= -\frac{\xi}{1+\alpha\xi} \frac{1}{1-\tau_t} \leq 0 \\ \frac{\partial \ln(s_t)}{\partial \tau_t} &= -\frac{1+\xi}{1+\alpha\xi} \frac{1}{1-\tau_t} \leq 0. \end{aligned}$$

We can also show that h_t is unambiguously decreasing in ϵ :

$$\begin{aligned} \frac{\partial \ln(h_t)}{\partial \epsilon} &= \frac{\xi}{1+\alpha\xi} \frac{\frac{1-\alpha}{\alpha} \frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon} \Gamma_{s,t}}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon} \Gamma_{s,t}} \frac{1}{1+\gamma_0\epsilon} \left((1+\gamma_0\epsilon) \frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} - \gamma_0 \right) \leq 0 \\ \frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon} &= \frac{\gamma_0}{1+\gamma_0\epsilon} \frac{\frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} (1+\theta\beta)}{1+\beta + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} (1+\theta\beta)}. \end{aligned}$$

³¹Recall informal workers are assumed to receive an endowment proportional to wages of formal workers when they retire. For tractability we define this as if they were making a labor supply decision. The relevant parameter determining the size of the endowment is χ .

However, the effect on s_t is harder to sign.

$$\frac{\partial \ln(s_t)}{\partial \epsilon} = \frac{1 + \xi}{\xi} \frac{\partial \ln(h_t)}{\partial \epsilon} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \epsilon}.$$

Replacing from the expressions above, and neglecting common factors, the sign of the derivative is given by the sign of

$$-\frac{1}{1 + \alpha\xi} \frac{\theta\beta(1 + \beta)}{1 + \beta + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} (1 + \theta\beta \frac{\xi}{1+\xi})} + 1 + \theta\beta.$$

This can be seen to be positive for all $\theta \in [0, 1]$. Note that the effect from γ_0 is equivalent to that of ϵ . We now consider the effect of future taxes on labor supply and savings.

$$\begin{aligned} \frac{\partial \ln(h_t)}{\partial \tau_{t+1}} &= \frac{\xi}{1 + \alpha\xi} \frac{\frac{1-\alpha}{\alpha} \frac{\theta\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon} \Gamma_{s,t}} \left(1 + \tau_{t+1} \frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}} \right) \geq 0 \\ \tau_{t+1} \frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}} &= -\frac{\frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}}{1 + \beta + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1, 0]. \end{aligned}$$

This shows that labor supply is increasing in τ_{t+1} . For savings the effect is:

$$\frac{\partial \ln(s_t)}{\partial \tau_{t+1}} = \frac{1 + \xi}{\xi} \frac{\partial \ln(h_t)}{\partial \tau_{t+1}} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \tau_{t+1}}.$$

As before the sign of this derivative can be shown to be the same as the sign of the following term:

$$\frac{1}{1 + \alpha\xi} \frac{\theta\beta(1 + \beta)}{1 + \beta + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0\epsilon} (1 + \theta\beta \frac{\xi}{1+\xi})} - (1 + \theta\beta).$$

This is basically the same term (with different sign) as before. Thus, for all values of θ and τ_{t+1} we have that an increase in τ_{t+1} reduces savings.

Taking the derivative with respect to θ , we see a similar condition as with the effect of τ_{t+1} , namely:

$$\begin{aligned} \frac{\partial \ln(h_t)}{\partial \theta} &= \frac{\xi}{1 + \alpha\xi} \frac{\frac{1-\alpha}{\alpha} \frac{\tau_{t+1}\Gamma_{s,t}}{1+\gamma_0\epsilon}}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau_{t+1}}{1+\gamma_0\epsilon} \Gamma_{s,t}} \left(1 + \theta \frac{\partial \ln(\Gamma_{s,t})}{\partial \theta} \right) \geq 0, \\ \theta \frac{\partial \ln(\Gamma_{s,t})}{\partial \theta} &= -\frac{\frac{1-\alpha}{\alpha} \frac{\tau_{t+1}\theta\beta}{1+\gamma_0\epsilon}}{1 + \beta + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1+\theta\beta)}{1+\gamma_0\epsilon}} \in (-1, 0]. \end{aligned}$$

This shows that labor supply is increasing in θ . For the savings rate, we have:

$$\frac{\partial \ln(s_t)}{\partial \theta} = \frac{1 + \xi}{\xi} \frac{\partial \ln(h_t)}{\partial \theta} + \frac{\partial \ln(\Gamma_{s,t})}{\partial \theta}.$$

Proceeding as before we find that this is unambiguously negative.³²

Part *ii.* follows from earlier derivations, specifically equation (A.3). Part *iii.* similarly follows from equation (A.8) with $B_{t+1} = \beta$ in the log case. Finally, part *iv.* follows straightforwardly, comparing (A.8) and (A.3): An average type i household has $\eta_i^{1+\xi}/X_i^\xi = \Gamma_h$. Thus, for $\tau_{t+1} = 0$ or $\theta = 1$, we simply have $s_{t,i}/s_t = \eta_i h_{t,i}/h_t$. The spread in savings rates increase in τ_{t+1} and decrease in θ in the sense that these parameter changes increase $s_{t,i}/s_t$ for $s_{t,i}/s_t > 1$ and decrease for $s_{t,i}/s_t < 1$.

Part *v.* The effects on $s_{t,0}$ and $s_{t,0}/s_t$ from various parameters come from differentiation of (A.9) with $B_{t+1}^0 = \beta$. Note that $s_{t,0}$ only depends on τ_t though the effect of taxes on s_t . Thus, it is straightforward to show that both $s_{t,0}$ and $s_{t,0}/s_t$ are increasing in τ_t .

The effects from τ_{t+1} and ϵ are given by:

$$\begin{aligned}\frac{\partial s_{t,0}}{\partial \tau_{t+1}} &= -\frac{1}{1+\beta} \frac{1-\alpha}{r_0 \alpha} \frac{s_t \epsilon}{1+\gamma_0 \epsilon} \left(1 + \tau_{t+1} \frac{\partial \ln(s_t)}{\partial \tau_{t+1}} \right) \leq 0, \\ \frac{\partial s_{t,0}}{\partial \epsilon} &= -\frac{1}{1+\beta} \frac{1-\alpha}{r_0 \alpha} \frac{s_t \tau_{t+1}}{1+\gamma_0 \epsilon} \left(\frac{1}{1+\gamma_0 \epsilon} + \epsilon \frac{\partial \ln(s_t)}{\partial \epsilon} \right) \leq 0.\end{aligned}$$

Where the first derivative can be signed since $\tau_{t+1} \frac{\partial \ln(s_t)}{\partial \tau_{t+1}} \geq -1$. It is straightforward to show that an increase in θ increases $s_{t,0}$.

To see the effect of τ_{t+1} on the ratio $s_{t,0}/s_t$ we have that:

$$\frac{\partial(s_{t,0}/s_t)}{\partial \tau_{t+1}} = -\frac{\beta}{1+\beta} \frac{\eta_0^{1+\xi}}{X_0^\xi (1+\xi)} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} \frac{1}{\Theta_{s,t}} \frac{\partial \ln(\Theta_{s,t})}{\partial \tau_{t+1}} - \frac{1}{1+\beta} \frac{1-\alpha}{r_0 \alpha} \frac{\epsilon}{1+\gamma_0 \epsilon},$$

where the derivative $\partial \ln(\Theta_{s,t})/\partial \tau_{t+1} = \partial \ln(s_t)/\partial \tau_{t+1}$ is defined above. We note that this may be negative or positive. For instance, if $\epsilon \approx 0$ an increase in future taxes increases this ratio. Conversely if $r_0 \approx 0$ then we would expect the opposite effect.

B Politico-economic equilibrium

B.1 Proof of proposition 2

As shown before, in equilibrium, all consumption functions are log-separable in the aggregate state s_{t-1}, h_{t-1} (see appendix A.1). In all the Θ -coefficient functions, the only *predetermined* variables that may enter the political problem as an endogenous state, is the distribution of savings $s_{t-1,i}/s_{t-1}$. However, the current distribution, $s_{t,i}/s_t$, is

³²The term whose sign determines this is slightly different

$$\frac{1}{1+\alpha\xi} \frac{1+\beta + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0 \epsilon}}{1+\beta + \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}}{1+\gamma_0 \epsilon} (1+\theta\beta \frac{\xi}{1+\xi})} - 1.$$

solely a function of parameters and the *current* policy. Thus, while $\{s_{t-1,i}/s_{t-1}\}_{i>0} \in z_t$, $\{s_{t,i}/s_t\}_{i>0}$ is independent of τ_t implying that

$$\frac{\partial z_{t+1}}{\partial \tau_t} = \frac{\partial \zeta_t}{\partial \tau_t} = 0,$$

where ζ_t is the law of motion for states in the political objective (eq. 9). Thus, we say that this distribution is not an *endogenous* state, as there are no strategic effects from changing τ_t on the future policy τ_{t+1} .

As there are no endogenous states, the politico-economic equilibrium is characterized by the first order condition

$$\begin{aligned} & \omega \left[\overbrace{\gamma_0 \frac{\partial \ln(\Theta_{\tilde{c}_2,t}^0)}{\partial \tau_t}}^{=\mathcal{E}_{2,t}^0} + \sum_i \gamma_i \overbrace{\frac{\partial \ln(\Theta_{c_2,t}^i)}{\partial \tau_t}}^{=\mathcal{E}_{2,t}^i} \right] \\ & + \nu_t \left\{ \underbrace{\gamma_0 \left(\frac{\partial \ln(\Theta_{\tilde{c}_1,t}^0)}{\partial \tau_t} + \beta \frac{\partial \ln(\Theta_{\tilde{c}_2,t+1}^0)}{\partial \tau_t} \right)}_{=\mathcal{E}_{1,t}^0} + \sum_i \gamma_i \underbrace{\left(\frac{\partial \ln(\Theta_{\tilde{c}_1,t}^i)}{\partial \tau_t} + \beta \frac{\partial \ln(\Theta_{c_2,t+1}^i)}{\partial \tau_t} \right)}_{=\mathcal{E}_{1,t}^i} \right\} \leq 0, \end{aligned}$$

with equality for an interior solution. We introduce the shorthand notation $\mathcal{E}_{x,t}^j$ to denote the marginal effect of taxes on the indirect utility of generation x of type j .

Using equilibrium consumption functions from appendix A.1, we have

$$\mathcal{E}_{2,t}^0 = \frac{(1-\alpha)\nu_t\epsilon \Theta_{h,t}^{1-\alpha} \left(1 + (1-\alpha)\tau_t \frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t} \right)}{\frac{(\chi\eta_0)^{1+\xi}}{(1+\xi)X_0^\xi} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{(1-\alpha)\nu_t\epsilon\tau_t}{1+\gamma_0\epsilon} \Theta_{h,t}^{1-\alpha}} \quad (\text{B.1})$$

$$\mathcal{E}_{2,t}^i = (1-\alpha) \frac{\partial \ln(\Theta_{h,t})}{\partial \tau_t} + \frac{\frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0\epsilon} \left(1 + \theta_t \left[\frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{\Gamma_h} - 1 \right] \right)}{\frac{s_{t-1}^i}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{\tau_t}{1+\gamma_0\epsilon} \left(1 + \theta_t \left[\frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{\Gamma_h} - 1 \right] \right)} \quad (\text{B.2})$$

$$\mathcal{E}_{1,t}^0 = -\frac{\beta}{1-\tau_t} \frac{\alpha(1+\xi)^2}{(1+\alpha\xi)^2} \quad (\text{B.3})$$

$$\mathcal{E}_{1,t}^i = -\frac{1}{1-\tau_t} \frac{1+\xi}{1+\alpha\xi} (1+\beta), \quad (\text{B.4})$$

It is straightforward to verify that all young households' indirect utility is decreasing in τ_t . For all retirees, there are opposing effects from adjusting the tax rate: The first term is negative and works through the labor response from current working households ($w_t h_t$ decreases with τ_t) on the pension contributions. The second term is positive and relates to the direct effect of higher pension benefits and the increase in the interest rate.

In an interior equilibrium it must be the case that retirees prefer positive taxes, as

the young unambiguously always support zero taxes. Then, increasing ω or decreasing ν_t naturally leads to higher equilibrium tax rates. In the corner with $\tau_t = 0$, there will be no effect from marginal changes in ω, ν_t . Thus $\partial\tau_t/\partial\omega \geq 0$ and $\partial\tau_t/\partial\nu_t \leq 0$. The sign in ν_t needs one more step, since ν_t also scales the universal pension relative to the endowment of informal retirees and so raises their support: with $\sigma_t^0 \equiv b_t^0/\tilde{c}_{2,t}^0$ the pension's share of their consumption, $\partial\mathcal{E}_{2,t}^0/\partial\nu_t = \mathcal{E}_{2,t}^0(1 - \sigma_t^0)/\nu_t$. Writing z_t for the left-hand side of the first order condition and using the condition itself to substitute for the young's terms, $\partial z_t/\partial\nu_t = -\frac{\omega}{\nu_t} \left[\gamma_0 \mathcal{E}_{2,t}^0 \sigma_t^0 + \sum_{i>0} \gamma_i \mathcal{E}_{2,t}^i \right]$, which is negative whenever the aggregate marginal support of formal retirees, $\sum_{i>0} \gamma_i \mathcal{E}_{2,t}^i$, is positive at the optimum, that is, unless the tax is sustained by informal retirees against formal ones.

To analyze the effect of income inequality on equilibrium taxes, without loss of generality we consider two types, H and L , with productivities above and below the average respectively. We increase the productivity for type H and that of type L such that

$$\frac{\partial\eta_L}{\partial\eta_H} = -\frac{\gamma_H \left(\frac{\eta_H}{X_H} \right)^\xi}{\gamma_L \left(\frac{\eta_L}{X_L} \right)^\xi},$$

which keeps the average labor supply and savings (and Γ_h) constant for a given path of policies. We note that for young generation, this does not alter the determinants of taxes as $\mathcal{E}_{1,t}^i$ is unaffected by this. The effect on relevant retirees $m = L, H$ are given by:

$$\frac{\partial\mathcal{E}_{2,t}^m}{\partial\eta_m} = -(1 + \xi) \left(\frac{\eta_m}{X_m} \right)^\xi \frac{1}{\Gamma_h} \frac{1 - \alpha}{\alpha} \frac{1}{1 + \gamma_0\epsilon} \frac{(1 - \theta) \left(1 + \frac{1 - \alpha}{\alpha} \frac{\tau_t}{1 + \gamma_0\epsilon} \frac{1}{1 + \beta} \right)}{\left(\frac{s_{t-1}^m}{s_{t-1}} + \frac{1 - \alpha}{\alpha} \frac{\tau_t}{1 + \gamma_0\epsilon} \left(1 + \theta_t \left[\frac{\eta_m^{1+\xi}}{X_m^\xi} \frac{1}{\Gamma_h} - 1 \right] \right) \right)^2}$$

This shows that relatively high income households have a weaker preference for taxes. Using this, it is now straightforward to show that $\partial(\gamma_L \mathcal{E}_{2,t}^L + \gamma_H \mathcal{E}_{2,t}^H)/\partial\eta_H > 0$. This is driven by the fact that the marginal utility of consumption is higher for low income households. Thus, higher inequality increases equilibrium taxes.

A similar argument can be applied to see the marginal effects on types H and L of an increase in θ :

$$\frac{\partial\mathcal{E}_{2,t}^m}{\partial\theta} = \frac{\frac{1 - \alpha}{\alpha} \frac{\tau_t}{1 + \gamma_0\epsilon} \left[\frac{\eta_m^{1+\xi}}{X_m^\xi} \frac{1}{\Gamma_h} - 1 \right]}{\frac{s_{t-1}^m}{s_{t-1}} + \frac{1 - \alpha}{\alpha} \frac{\tau_t}{1 + \gamma_0\epsilon} \left(1 + \theta_t \left[\frac{\eta_m^{1+\xi}}{X_m^\xi} \frac{1}{\Gamma_h} - 1 \right] \right)} \frac{s_{t-1}^m/s_{t-1}}{\frac{s_{t-1}^m}{s_{t-1}} + \frac{1 - \alpha}{\alpha} \frac{\tau_t}{1 + \gamma_0\epsilon} \left(1 + \theta_t \left[\frac{\eta_m^{1+\xi}}{X_m^\xi} \frac{1}{\Gamma_h} - 1 \right] \right)}$$

Thus, an increase in θ increases the support for taxes among the high income households but weakens it among the poor. For the same reasons as above, namely the higher marginal utility of poor households, higher θ then lowers equilibrium tax rates.

The effect of ϵ works through retirees alone, since $\mathcal{E}_{1,t}^0$ and $\mathcal{E}_{1,t}^i$ do not depend on ϵ . Write $\sigma_t^j \equiv b_t^j/\tilde{c}_{2,t}^j$ for the pension's share of the consumption of a retiree of type j , as σ_t^0

above, and $\phi_t \equiv 1 + (1 - \alpha)\tau_t \partial \ln(\Theta_{h,t})/\partial \tau_t$ for the marginal revenue of the tax relative to its base. The expressions above are then $\mathcal{E}_{2,t}^0 = \phi_t \sigma_t^0/\tau_t$ and $\mathcal{E}_{2,t}^i = (\phi_t \sigma_t^i - (1 - \phi_t)(1 - \sigma_t^i))/\tau_t$: a unit of tax raises every pension by ϕ_t/τ_t in proportion, and with log utility a retiree's support is that proportion times the pension's share of consumption, less, for formal retirees, the fall in the interest rate on their savings. Benefits are shares of revenue $\nu_t w_t h_t \tau_t$, $\epsilon/(1 + \gamma_0 \epsilon)$ per informal retiree and $(1 + \theta_t[y_i - 1])/(1 + \gamma_0 \epsilon)$ per formal retiree of type i with $y_i \equiv \frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{\Gamma_h}$, so a higher ϵ raises b_t^0 by $1/(\epsilon(1 + \gamma_0 \epsilon))$ in proportion and lowers every b_t^i by $\gamma_0/(1 + \gamma_0 \epsilon)$ in proportion, a pure transfer since there are γ_0 informal retirees per formal one. With $s_{t-1,i}/s_{t-1}$ and $\Theta_{h,t}$ held fixed as in the comparative statics above, ϵ enters the first order condition only through the shares, and a share responds to a proportional change in the benefit by $\partial \sigma_t^j / \partial \ln b_t^j = \sigma_t^j(1 - \sigma_t^j)$, since the rest of consumption does not move. Hence

$$\frac{\partial z_t}{\partial \epsilon} = \frac{\omega \gamma_0}{\tau_t(1 + \gamma_0 \epsilon)} \left[\frac{\phi_t}{\epsilon} \sigma_t^0(1 - \sigma_t^0) - \sum_{i>0} \gamma_i \sigma_t^i(1 - \sigma_t^i) \right].$$

At an interior equilibrium z_t is decreasing in τ_t , so taxes rise with ϵ if and only if the bracket is positive,

$$\frac{\phi_t}{\epsilon} \sigma_t^0(1 - \sigma_t^0) > \sum_{i>0} \gamma_i \sigma_t^i(1 - \sigma_t^i). \quad (\text{B.5})$$

The share γ_0 multiplies the whole expression, because the loss of formal retirees is the gain of informal retirees times their number, and enters the bracket only through the consumption levels, so it does not determine the sign. The factor $\sigma(1 - \sigma)$ is zero for a retiree without a pension and for one who lives on the pension alone, whose consumption and marginal benefit a higher ϵ raises in the same proportion, and is largest at one half. The factor $1/\epsilon$ is the asymmetry of the transfer: a unit of ϵ is a large proportional change of a small universal pension and a small one of the formal pensions. The condition therefore holds when informal retirees are poorer than formal ones and the pension is a comparable share of their consumption, since a comparable share of a smaller consumption is a smaller pension, and fails when they live on the pension alone or are much richer than formal retirees.

B.2 With log-GHH preferences and informal savings

The introduction of $s_{t-1,0}/s_{t-1}$ as an endogenous state variable complicates the model because this state variable not only depends on past choices (through the effect of τ_{t-1} on $\Theta_{s,t-1}$), but is also forward looking ($s_{t-1,0}/s_{t-1}$ depends explicitly on τ_t). To deal with this, let $dx_t/d\tau_t$ denote the effect in politico-economic equilibrium of changing the tax rate τ_t on the variable x_t accounting for the fact that a change in τ_t also changes

the state $s_{t,0}/s_t$ which through the continuation policy affects the choice of τ_{t+1} as well. Specifically, for the state variable $s_{t,0}/s_t$ we have:

$$\frac{d(s_{t,0}/s_t)}{d\tau_t} = \frac{\partial(s_{t,0}/s_t)}{\partial\tau_t} + \frac{\partial(s_{t,0}/s_t)}{\partial\tau_{t+1}} \frac{d\tau^{t+1}}{d(s_{t,0}/s_t)} \frac{d(s_{t,0}/s_t)}{d\tau_t},$$

where the partial derivatives of $s_{t,0}/s_t$ captures how the state reacts to policy in economic equilibrium and $\tau^{t+1}(s_{t,0}/s_t)$ is the continuation policy that details how policy in $t+1$ adjusts when the endogenous state changes. When considering a marginal change in τ_t , the policymaker takes into account how this propagates via economic equilibrium and continuation policy functions into future periods in order to summarize the total impact on current relevant variables. The technical documentation derives a recursive structure for all the relevant derivatives required to evaluate the marginal effect of τ_t on the indirect utility of households (**BergGE26doc**).

With the endogenous state $s_{t-1,0}/s_{t-1}$, the politico-economic equilibrium is now characterized by the first order condition:

$$\omega \left[\sum_{i \geq 0} \gamma_i \overbrace{\frac{d \ln(c_{2,t}^i)}{d\tau_t}}^{=\mathcal{E}_{2,t}^i} \right] + \nu_t \left\{ \sum_{i \geq 0} \gamma_i \underbrace{\left(\frac{d \ln(\tilde{c}_{1,t}^i)}{d\tau_t} + \beta \frac{d \ln(c_{2,t+1}^i)}{d\tau_t} \right)}_{=\mathcal{E}_{1,t}^i} \right\} \leq 0,$$

Using equilibrium consumption functions from appendix A.1, we have the derivatives of indirect utility for retired households:

$$\mathcal{E}_{2,t}^0 = (1 - \alpha) \frac{d \ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{r_0\alpha} \frac{\epsilon}{1+\gamma_0\epsilon}}{\frac{s_{t-1,0}}{s_{t-1}} + \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_t}{1+\gamma_0\epsilon}} \quad (\text{B.6})$$

$$\mathcal{E}_{2,t}^i = (1 - \alpha) \frac{d \ln(h_t)}{d\tau_t} + \frac{\frac{1-\alpha}{\alpha} \frac{1}{1+\gamma_0\epsilon} \left(1 + \theta_t \left[\frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{\Gamma_h} - 1 \right] \right)}{\frac{s_{t-1,i}}{s_{t-1}} + \frac{1-\alpha}{\alpha} \frac{\tau_t}{1+\gamma_0\epsilon} \left(1 + \theta \left[\frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{\Gamma_h} - 1 \right] \right)}. \quad (\text{B.7})$$

We note that the marginal effect for formal retirees is the same as when continuation policies were independent of state variables. The only difference is that the change in the aggregate labor supply, $\frac{d \ln(h_t)}{d\tau_t}$, now includes strategic effects through future policies.

For the formal, working households, we can use the Euler equation to write the indirect utility as $\ln(\tilde{c}_{1,t}^i) (1 + \beta) + \beta \ln(\beta R_{t+1})$. The marginal effect on indirect utility then consists of the two terms:

$$\mathcal{E}_{1,t}^i = (1 + \beta) \frac{d \ln(\tilde{c}_{1,t}^i)}{d\tau_t} + \beta \frac{d \ln(R_{t+1})}{d\tau_t}$$

$$\begin{aligned}\frac{d \ln(\tilde{c}_{1,t}^i)}{d\tau_t} &= \frac{1+\xi}{\xi} \frac{d \ln(h_t)}{d\tau_t} + \frac{\Gamma_{s,t} \frac{1-\alpha}{\alpha} \frac{1-\theta_{t+1}}{1+\gamma_0\epsilon} \left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1} \frac{d \ln(\Gamma_{s,t})}{d\tau_t} \right)}{\frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{1+\xi} + \Gamma_{s,t} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}} \\ \frac{d \ln(R_{t+1})}{d\tau_t} &= (1-\alpha) \left(\frac{d \ln(h_{t+1})}{d \ln(s_t)} - 1 \right) \frac{d \ln(s_t)}{d\tau_t}.\end{aligned}$$

where we use the shorthand $d\tau_{t+1}/d\tau_t = \frac{d\tau^{t+1}}{d(s_t^0/s_t)} \frac{d(s_t^0/s_t)}{d\tau_t}$ to define the total effect of a change in the tax rate on future taxes. The technical documentation gives the computation of these derivatives in the numerical application (**BergGE26doc**).

For the informal, working households, a similar Euler condition applies such that we can write indirect utility as $\ln(\tilde{c}_{1,t}^0) (1+\beta) + \beta \ln(\beta R_{t+1}^0)$. The marginal effect on indirect utility is then characterized by

$$\begin{aligned}\mathcal{E}_{1,t}^0 &= (1+\beta) \frac{d \ln(\tilde{c}_{1,t}^0)}{d\tau_t} + \beta \frac{d \ln(R_{t+1}^0)}{d\tau_t} \\ \frac{d \ln(\tilde{c}_{1,t}^0)}{d\tau_t} &= \frac{\frac{1-\alpha}{r_0\alpha} \frac{\epsilon}{1+\gamma_0\epsilon} \Theta_{s,t} \left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1} \frac{d \ln(s_t)}{d\tau_t} \right)}{\frac{\eta_0^{1+\xi}}{X_0^\xi} \frac{1}{1+\xi} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} + \frac{1-\alpha}{r_0\alpha} \frac{\epsilon\tau_{t+1}}{1+\gamma_0\epsilon} \Theta_{s,t}} \\ \frac{d \ln(R_{t+1}^0)}{d\tau_t} &= (1-\alpha) \left(\frac{d \ln(h_{t+1})}{d \ln(s_t)} - 1 \right) \frac{d \ln(s_t)}{d\tau_t}\end{aligned}$$

B.3 With CRRA-GHH preferences and informal savings

With CRRA preferences it is no longer the case that future policy is independent of the capital stock. But conditional on this state variable, with GHH preferences it is still the case that the distribution of savings for formal workers, $\{s_{t-1,i}/s_{t-1}\}_{i>0}$, remains a function of s_{t-1} and τ_t and is thus not an independent state variable. Thus, the policy function can be written as a function of two endogenous states: s_{t-1} and $s_{t-1,0}/s_{t-1}$.

Compared to the log-GHH case, we adjust the definition of the politico-economic equilibria derivative $dx_t/d\tau_t$ to include the effect through both states such that

$$\frac{dx_t}{d\tau_t} = \frac{\partial x_t}{\partial \tau_t} + \frac{\partial x_t}{\partial \tau_{t+1}} \left(\frac{\partial \tau^{t+1}}{\partial s_t} \frac{ds_t}{d\tau_t} + \frac{\partial \tau^{t+1}}{\partial (s_{t,0}/s_t)} \frac{d(s_{t,0}/s_t)}{d\tau_t} \right).$$

The first order condition identifying the politico-economic choice τ_t is now now given by

$$\omega \sum_{j \geq 0} \gamma_j \underbrace{\left(c_{2,t}^j \right)^{\frac{\rho-1}{\rho}} \frac{d \ln(c_{2,t}^j)}{d\tau_t}}_{\equiv \mathcal{E}_{2,t}^j} + \nu_t \sum_{j \geq 0} \gamma_j \underbrace{\left(\left(\tilde{c}_{1,t}^j \right)^{\frac{\rho-1}{\rho}} \frac{d \ln(\tilde{c}_{1,t}^j)}{d\tau_t} + \beta \left(c_{2,t+1}^j \right)^{\frac{\rho-1}{\rho}} \frac{d \ln(c_{2,t+1}^j)}{d\tau_t} \right)}_{\equiv \mathcal{E}_{1,t}^j} \leq 0,$$

The log-derivatives $d \ln (c_{2,t}^j) / d\tau_t$ follow from the formulas for the log-GHH case and consumption levels $c_{2,t}^j$ depend on the state s_{t-1} and are given in appendix A.1.

For formal working households, we use the Euler equation to write indirect utility as

$$\begin{aligned} v_{1,t}^i &= (1 + B_{t+1}) \frac{\left(\hat{c}_{1,t}^i\right)^{1-1/\rho}}{1 - 1/\rho} = \frac{\left(\hat{c}_{1,t}^i\right)^{1-1/\rho}}{1 - 1/\rho} \\ \hat{c}_{1,t}^i &= \left(\frac{h_t}{\Gamma_h}\right)^{\frac{1+\xi}{\xi}} (1 + B_{t+1})^{\frac{1}{\rho-1}} \mathcal{C}_{1,t}^i \\ \mathcal{C}_{1,t}^i &= \frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{1+\xi} + \Gamma_{s,t} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon} \end{aligned}$$

The marginal effect of taxes can then be summed up by

$$\frac{dv_{1,t}^i}{d\tau_t} = \left(\hat{c}_{1,t}^i\right)^{1-1/\rho} \frac{d \ln (\hat{c}_{1,t}^i)}{d\tau_t} \quad (\text{B.8})$$

$$\begin{aligned} \frac{d \ln (\hat{c}_{1,t}^i)}{d\tau_t} &= \frac{1+\xi}{\xi} \frac{d \ln (h_t)}{d\tau_t} + \frac{B_{t+1}}{1+B_{t+1}} (1-\alpha) \left(\frac{d \ln (h_{t+1})}{d \ln (s_t)} - 1 \right) \frac{d \ln (s_t)}{d\tau_t} \\ &\quad + \frac{\Gamma_{s,t} \frac{1-\alpha}{\alpha} \frac{1-\theta}{1+\gamma_0\epsilon} \left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1} \frac{d \ln (\Gamma_{s,t})}{d\tau_t} \right)}{\frac{\eta_i^{1+\xi}}{X_i^\xi} \frac{1}{1+\xi} + \Gamma_{s,t} \frac{1-\alpha}{\alpha} \frac{\tau_{t+1}(1-\theta)}{1+\gamma_0\epsilon}} \end{aligned} \quad (\text{B.9})$$

For the young informal households, we can employ the Euler equation in a similar way and write:

$$\begin{aligned} v_{1,t}^0 &= \frac{\left(\hat{c}_{1,t}^0(1 + B_{t+1}^0)^{\frac{\rho}{\rho-1}}\right)^{1-1/\rho}}{1 - 1/\rho} = \frac{\left(\hat{c}_{1,t}^0\right)^{1-1/\rho}}{1 - 1/\rho} \\ \hat{c}_{1,t}^0 &= \left(\frac{s_{t-1}}{\nu_t}\right)^{\sigma_s} (1 + B_{t+1}^0)^{\frac{1}{\rho-1}} \mathcal{C}_{1,t}^0 \\ \mathcal{C}_{1,t}^0 &= \frac{\eta_0^{1+\xi}}{X_0^\xi} \frac{1}{1+\xi} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} + \Theta_{s,t} \frac{1-\alpha}{r_0\alpha} \frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon} \end{aligned}$$

We then have that:

$$\frac{dv_{1,t}^0}{d\tau_t} = \left(\hat{c}_{1,t}^0\right)^{1-1/\rho} \frac{d \ln (\hat{c}_{1,t}^0)}{d\tau_t} \quad (\text{B.10})$$

$$\begin{aligned} \frac{d \ln (\hat{c}_{1,t}^0)}{d\tau_t} &= \frac{B_{t+1}^0}{1+B_{t+1}^0} (1-\alpha) \left(\frac{d \ln (h_{t+1})}{d \ln (s_t)} - 1 \right) \frac{d \ln (s_t)}{d\tau_t} + \frac{\Theta_{s,t} \frac{1-\alpha}{r_0\alpha} \frac{\epsilon}{1+\gamma_0\epsilon} \left(\frac{d\tau_{t+1}}{d\tau_t} + \tau_{t+1} \frac{d \ln (\Theta_{s,t})}{d\tau_t} \right)}{\frac{\eta_0^{1+\xi}}{X_0^\xi} \frac{1}{1+\xi} \left(\frac{1-\alpha}{\Gamma_h^\alpha} \right)^{\frac{1+\xi}{1+\alpha\xi}} + \Theta_{s,t} \frac{1-\alpha}{r_0\alpha} \frac{\tau_{t+1}\epsilon}{1+\gamma_0\epsilon}} \end{aligned} \quad (\text{B.11})$$

B.4 The Beveridgean corner

We prove the claim of section 7.1. Without informal households, (8) and (A.3) give a retiree of type i the benefit $\nu_t w_t h_t \tau_t (1 + \theta_t [y_i - 1])$, where y_i of (15) equals $\eta_i h_{t-1,i}/h_{t-1}$, the labor income of type i relative to the mean.

Proposition 3. *Assume log-GHH preferences, no informal households and no deadweight cost of redistribution (section 7.2), and let the design θ_t that applies to the benefits paid at t be chosen at t together with τ_t . Let the propensities to vote be positive and not rise with relative income, $\mu_i \leq \mu_j$ whenever $y_i > y_j$, equal propensities included, and let relative incomes not all be equal. Then, for every $\omega > 0$ and $\tau_t > 0$, the derivative (15) is negative at every $\theta_t \in [0, 1]$ at which every retiree's consumption is positive, and the design chosen together with a positive tax is the Beveridgean corner, $\theta_t = 0$. This holds for all values of $\alpha, \beta, \xi, \nu_t, \gamma_i, \eta_i$ and X_i , and whatever tax and design current retirees expected when they saved.*

Proof of proposition 3. By (11) the labor supply and the savings of the young at t depend on the design that will apply to their own benefits, θ_{t+1} , and not on θ_t , so θ_t enters no state at $t + 1$ and no later decision. It enters $\mathcal{W}_t$ only through the consumption of current retirees (appendix A.1, with $\gamma_0 = 0$),

$$c_{2,t}^i = \alpha \nu_t h_t^{1-\alpha} \left(\frac{s_{t-1}^i}{\nu_t} \right)^\alpha D_i, \quad D_i \equiv \frac{s_{t-1}^i}{s_{t-1}} + \frac{1-\alpha}{\alpha} \tau_t (1 + \theta_t [y_i - 1]), \quad (\text{B.12})$$

where the first factor is common to all types and independent of θ_t . From (14),

$$\frac{\partial \mathcal{W}_t}{\partial \theta_t} = \omega \sum_i \gamma_i \mu_i \frac{\partial \ln c_{2,t}^i}{\partial \theta_t} = \omega \frac{1-\alpha}{\alpha} \tau_t \sum_i \gamma_i (y_i - 1) \frac{\mu_i}{D_i},$$

which is (15). We sign the sum in three steps.

First, relative incomes average to one. By the definition of Γ_h in (A.3), $\sum_i \gamma_i y_i = 1$, and since $\sum_i \gamma_i = 1$ (section 2.1), $\sum_i \gamma_i (y_i - 1) = 0$: the design moves a given total between retirees below and above mean income.

Second, D_i rises with y_i . The savings of current retirees were chosen at $t - 1$. By part iii of proposition 1, that is (A.8) with $B_{t+1} = \beta$, in which the tax and the design are those that apply to the benefits of the generation that saves,

$$\frac{s_{t-1}^i}{s_{t-1}} = y_i + \frac{1-\alpha}{\alpha} \frac{\tau_t^e (1 - \theta_t^e)}{1 + \beta} (y_i - 1),$$

where τ_t^e and θ_t^e are the tax and the design that generation $t - 1$ expected for period t . Whatever these were, the slope in y_i is at least one, since $\tau_t^e \geq 0$ and $\theta_t^e \leq 1$. The second

term of D_i has slope $\frac{1-\alpha}{\alpha}\tau_t\theta_t \geq 0$, so D_i rises with y_i with a slope of at least one, and by (B.12) it is positive wherever every retiree's consumption is.

Third, Chebyshev's sum inequality: for weights $\gamma_i \geq 0$ that sum to one,

$$\sum_i \gamma_i a_i b_i - \sum_i \gamma_i a_i \sum_j \gamma_j b_j = \frac{1}{2} \sum_{i,j} \gamma_i \gamma_j (a_i - a_j) (b_i - b_j),$$

which is non-positive when a_i and b_i are ordered in opposite directions. Let $a_i = y_i - 1$ and $b_i = \mu_i/D_i$. If $y_i > y_j$, then $\mu_i \leq \mu_j$ and $D_i > D_j > 0$, so $b_i < b_j$, and every term with $y_i \neq y_j$ is negative. Since $\sum_i \gamma_i a_i = 0$ by the first step and relative incomes are not all equal, $\sum_i \gamma_i (y_i - 1) \mu_i/D_i < 0$, and so is the derivative.

Finally, D_i falls with θ_t only for types below mean income, and $D_i \geq 1$ for the others. The designs at which every retiree's consumption is positive therefore form an interval that starts at $\theta_t = 0$ whenever there are any, $\mathcal{W}_t$ is strictly decreasing on it, and the design chosen with any positive tax is $\theta_t = 0$.

The proof uses the propensities to vote only through the ratios μ_i/D_i , which order the types as $\mu_i/c_{2,t}^i$ does. The derivative is therefore negative at every design at which propensities to vote rise with relative income, if at all, by proportionally less than consumption in retirement, $\mu_i/\mu_j < c_{2,t}^i/c_{2,t}^j$ whenever $y_i > y_j$, and a design other than the Beveridgean corner requires propensities to vote that, for some pair of types and at some design, rise at least as fast as consumption. The argument also holds under CRRA-GHH preferences at every intertemporal elasticity: the design still enters only the consumption of current retirees, whose marginal utility is $D_i^{-1/\rho}$ times a factor common to all types, and relative savings keep the form of (A.8) with a discount term B_t common to all types, so $\mu_i D_i^{-1/\rho}$ takes the place of μ_i/D_i .

C Terminal states

We consider two different terminal states of the model: A finite horizon version where the economy ends after $t = T$ and a steady state version. The paper reports the finite-horizon solution, in periods of thirty years from 1960 to $T = 2200$ for the rich economies and from 1920 to $T = 2190$ for Argentina, with population growth held at its last projected rate after 2080 (2070 for Argentina).

Finite Horizon. In the finite horizon, consumption functions and indirect utility in economic equilibrium are all identical to years prior $t < T$. For the young generations at time T , we simply say that

$$u_{T,j} = u_j(c_{1,T}^j, h_{T,j}).$$

In this case, the economic equilibrium reduces to the general formulation only with $\beta_j = \tau_{t+1} = s_t = s_t^j = 0$. Specifically:

$$h_t = (\Gamma_h)^{\frac{1}{1+\alpha\xi}} ((1-\alpha)(1-\tau_t))^{\frac{\xi}{1+\alpha\xi}} \left(\frac{s_{t-1}}{\nu_t} \right)^{\frac{\alpha\xi}{1+\alpha\xi}} \quad (\text{C.1})$$

$$h_{t,i} = h_t \frac{(\eta_i/X_i)^\xi}{\Gamma_h} \quad (\text{C.2})$$

$$\tilde{c}_{1,t}^i = \frac{1}{1+\xi} \frac{\eta_i^{1+\xi}}{X_i^\xi} \left(\frac{h_t}{\Gamma_h} \right)^{\frac{1+\xi}{\xi}} \quad (\text{C.3})$$

$$\tilde{c}_{1,t}^0 = \frac{1}{1+\xi} \frac{\eta_0^{1+\xi}}{X_0^\xi} (w_t^0)^{1+\xi}. \quad (\text{C.4})$$

For the retired generation, the relevant consumption functions are unchanged.

For the politico-economic equilibrium, the marginal effects of τ_t is still equivalent for retirees (to the case where $t < T$). For working households, the effects are much simpler, namely:

$$\mathcal{E}_{1,t}^i = (\tilde{c}_{1,t}^i)^{1-1/\rho} \frac{1+\xi}{\xi} \frac{\partial \ln(h_t)}{\partial \tau_t} \quad (\text{C.5})$$

$$\mathcal{E}_{1,t}^0 = 0 \quad (\text{C.6})$$

Steady state, infinite horizon. In steady state, we combine equations for aggregate labor supply and savings states to write:

$$s^* = \Gamma_h^{1+\xi} \left[\left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s} \right)^{1+\xi} \frac{\Gamma_s^{1+\alpha\xi}}{\nu^{\alpha(1+\xi)}} \right]^{\frac{1}{1-\alpha}} \quad (\text{C.7})$$

$$\Gamma_s = \frac{1}{1+\xi} \frac{\Gamma_h B}{1+B + \frac{1-\alpha}{\alpha} \frac{\tau}{1+\gamma_0\epsilon} (1+\theta B)} \quad (\text{C.8})$$

$$B^i = (\beta)^\rho \left(\frac{\alpha}{p} \left(\frac{s^*}{\nu h^*} \right)^{\alpha-1} \right)^{\rho-1}$$

$$\frac{s^*}{\nu h^*} = \frac{(s^*)^{\frac{1}{1+\xi}} \Gamma_s^{\frac{\xi}{1+\xi}}}{\nu \Gamma_h} = \left(\frac{(1-\alpha)(1-\tau)}{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s} \frac{\Gamma_s}{\nu} \right)^{\frac{1}{1-\alpha}} \quad (\text{C.9})$$

We can use this to write the discount function as

$$B = (\beta)^\rho \left(\alpha \frac{\Gamma_h - \frac{1-\alpha}{\alpha} \frac{\theta\tau}{1+\gamma_0\epsilon} \Gamma_s}{(1-\alpha)(1-\tau)} \frac{\nu}{\Gamma_s} \right)^{\rho-1} \quad (\text{C.10})$$

D Calibration – Argentina

This appendix holds the data, the identifying formulas and the sources behind table 1, which section 5 summarizes.

D.1 The household survey

The Encuesta Permanente de Hogares (EPH) of the Instituto Nacional de Estadística y Censos (INDEC) is Argentina’s main household survey of employment, hours and income, and the basis of its official labor market statistics. It is a continuous quarterly survey of the urban agglomerates, 31 metropolitan areas at present, in which households are drawn by a stratified multi-stage design. We use the last two quarters of 2006, a phase of sustained expansion after the recovery from the 2001–2002 crisis, at the start of the expansion of coverage under the amnesties (figure 5.1) and before the Great Recession. The survey asks respondents whether they contribute to social security, which separates formal from informal workers, and we keep those aged 25 to 60.

D.2 The pension system before and after the reform

Before the tax amnesties, retirees over the age of 70 who had made contributions for a minimum of 10 years were eligible for basic pension coverage. This coverage consisted of 70% of the minimum pension and a proportional amount from other contributions.³³ We calculated the pre-reform ϵ

$$\epsilon = 0.7 \left(1 - \theta + \theta \frac{\eta^1 h_t^1}{h_t} \right) \frac{\beta^{5/30} \frac{LE_{men} - 5}{LE_{men}} + \beta^{10/30} \frac{LE_{women} - 10}{LE_{women}}}{2},$$

where $LE_{men} = 14.45$ and $LE_{women} = 22.55$ are the expected life expectancy for men and women, respectively, from their retirement ages (women retire at age 60, men at 65), such that the formula above corrects for the basic pension being received 5 or 10 years after the formal retirement age.³⁴ We set γ_0 so that informal households are 32% of all households, the share of the population over 65 without a pension before the reform, in 2004 (**Roffman**); since the formal shares sum to one, $\gamma_0 = 0.32/0.68 = 0.47$. To identify the Bismarckian incentives, we use OECD data on the ratio of replacement rates, RR ,

³³The basic pensions was set by law 24347 of year 1995 at 70% of the first pillar, denoted “prestación básica universal”, plus the corresponding fully-funded or pay-as-you-go benefit. For simplicity we assume the latter on average was also 70% of the benefits received by workers that had at least 30 years of contributions. Thus, basic pensions were 70% of the benefits received by the least productive formal workers.

³⁴To estimate life expectancy at retirement age for Argentina we conduct a regression analysis using data for men and women’s life expectancies at birth and at age 65 for 40 OECD countries for the year 2000. We then use the fitted regression and data from the World Bank on male and female life expectancy at birth in Argentina in 2000.

at half mean and mean wages for males and females.³⁵ Specifically, we adjust θ to ensure that

$$RR \equiv \frac{\theta + (1 - \theta) \frac{h_t}{\eta_t^2 h_t^2}}{\theta + (1 - \theta) \frac{h_t}{\eta_t^1 h_t^1}} = \frac{0.678}{0.803},$$

where the superscripts 1 and 2 denote the quartiles whose mean incomes are closest to half the formal average and to the average, the first and the third, at 0.47 and 1.00 of the average in the survey.

D.3 Technology, demographics, and the aggregate targets

The capital income share is that of the formal sector. In the model, informal households produce outside Y_t and K_t , and national accounts book the output of unincorporated household enterprises, where most informal work sits, as mixed income, which a share computed as one minus the wage bill over GDP counts entirely as capital (**Gollina1**). The economy-wide share for Argentina is 0.43 (**Frankema**). If a share y of GDP is informal mixed income, the formal-sector share is $(0.43 - y)/(1 - y)$. Informal earnings of about 5% of GDP, as in the household survey, give 0.40, while shadow-economy estimates of 20–25% of GDP give 0.24 to 0.29. We take $\alpha = 0.35$, which also puts Argentina on the same footing as the Gollin-adjusted shares we use for the rich economies in section 6. We set the Frisch elasticity of labor supply to $\xi = 0.30$. To calibrate β , we target the capital-output ratio. A model period is thirty years and capital depreciates fully between periods, so the capital stock in place at t is the previous generation's savings, $K_t = s_{t-1}/\nu_t$, and with Cobb–Douglas production $K_t/Y_t = \alpha/R_t$ — a stock relative to one period, i.e. thirty years, of output. Thirty times that ratio is the capital-output ratio as the national accounts report it. In our model with $K_t/Y_t = \alpha/R_t$ this is essentially equivalent to targeting the thirty year interest rate, which for Argentina is not credibly measured. We target $K/Y = 3.23$, Argentina's ratio in 2010.³⁶ We calibrate β and ϵ numerically, since β depends on the politico-economic equilibrium path for savings and labor supply and ϵ depends on β .

To construct the time series for ν_t we follow **mkvss08** and use 30-year gross population growth rate and projections from census data. Finally, we use social security spending

³⁵In a pure Bismarckian system the replacement rate would be the same at all wage levels, such that $RR = 1$, while in a pure Beveridgean it would double if we halve wages, implying $RR = 0.5$. Note that we combine men and women since we do not have data separately on the increase in coverage or government spending by gender.

³⁶Capital stock and GDP at constant national prices (**rna**, **rgdpna**) from the Penn World Table 11.0 (**pwt**), retrieved through FRED. We read it at the calibration year, where the tax rate, the replacement-rate ratio, the coverage share and the household survey are all measured, rather than averaging. Correcting for the shadow economy would raise the capital-output target, since the Penn World Table stock is economy-wide and the informal sector is labor-intensive. We keep the measured ratio, because no source measures an informal capital share and the stock also includes housing, which cuts the other way.

before the 2001 crisis of 7.1% of GDP,³⁷ to calibrate ω . Since the payroll tax falls on formal labor income, which is a share $1 - \alpha$ of formal output, spending of 7.1% of GDP corresponds to an equivalent social security tax rate of $\tau_{2010} = 0.071/(1 - \alpha) = 0.109$.

D.4 Productivity and hours

We use η_j and X_j to match the income distribution across formal households and the income and working hours of informal households. Specifically, let z_j^η and z_j^x denote income and hours worked of group j relative to the population-weighted average over the formal households, so that $\sum_{i>0} \gamma_i z_i^\eta = \sum_{i>0} \gamma_i z_i^x = 1$; informal households are measured against the same formal average. With labor supply $h_t^i/h_t = (\eta_i/X_i)^\xi / \sum_{j>0} \gamma_j \eta_j^{1+\xi}/X_j^\xi$ for formal households (cf. proposition 1) and $h_t^0 = (\eta_0 w_t^0/X_0)^\xi$ for informal households, where w_t^0 is the informal wage, we solve the system

$$\begin{aligned} z_i^\eta &= \frac{w_t(1 - \tau_t)h_t^i \eta_i}{w_t(1 - \tau_t)h_t} = \frac{\eta_i^{1+\xi}/X_i^\xi}{\sum_{j>0} \gamma_j \eta_j^{1+\xi}/X_j^\xi}, & i > 0, \\ z_i^x &= \frac{h_t^i}{\sum_{j>0} \gamma_j h_t^j} = \frac{(\eta_i/X_i)^\xi}{\sum_{j>0} \gamma_j (\eta_j/X_j)^\xi}, & i > 0, \\ z_0^\eta &= \frac{w_t^0 h_t^0 \eta_0}{w_t(1 - \tau_t)h_t}, & z_0^x &= \frac{h_t^0}{\sum_{j>0} \gamma_j h_t^j}. \end{aligned} \tag{D.1}$$

Because only relative income and hours are observed, both the level of efficiency units and the unit in which hours are measured are free. We spend the first on the normalization $\sum_{i>0} \gamma_i \eta_i^{1+\xi}/X_i^\xi = 1$, and take the taste for leisure to be common across the formal groups, $X_i = X$ for $i > 0$. The formal income conditions then invert in closed form, $\eta_i = (z_i^\eta)^{1/(1+\xi)} X^{\xi/(1+\xi)}$, and they satisfy the normalization at any X , so the level of X is what fixes the unit in which hours are counted. We pin it down by requiring average formal hours in the baseline year, $\sum_{i>0} \gamma_i h_t^i$, to equal the average workweek of formal workers in the survey, 42.5 hours expressed as a share of discretionary time. Relative hours across the formal quartiles are then a prediction rather than a target, rising with relative income at elasticity $\xi/(1 + \xi)$, and section 5 compares the prediction with the survey. The aggregate h_t weights hours by productivity and is not average formal hours; the two differ by the factor $\sum_{i>0} \gamma_i (\eta_i/X_i)^\xi$, which the informal hours condition in (D.1) carries. That condition and the informal income condition pin down η_0 and X_0 given the equilibrium h_t in the baseline year, and we solve for them jointly with β and ω at $X = 1$. The level of X then follows in one step from the solved hours, and η_0 and X_0 are re-derived at it. That last step leaves the equilibrium untouched, since the informal

³⁷Total public pension spending, national and provincial schemes, 7.0% and 7.2% of GDP in 2000 and 2001 (**CetrangoloG**). We use pre-crisis spending as it was relatively stable during that time, and after 2010 contributive pensions returned to their pre-crisis levels. Following the 2001 crisis, pension spending was reduced, to 5.2% of GDP in 2005, by only allowing minimum pensions to adjust for inflation.

household reaches every aggregate through $\eta_0^{1+\xi}/X_0^\xi$, which the unit of hours does not move.

D.5 The vector- X_i calibration

The alternative to a common taste for leisure treats the survey's relative hours as data at every point of the formal distribution: with both z_i^η and z_i^x imposed, (D.1) identifies one X_i per quartile, and the level of hours is then not identified. Since the productivity distribution reaches every aggregate only through $\sum_{i>0} \gamma_i \eta_i^{1+\xi}/X_i^\xi$, which both calibrations normalize to one, β , ω and every equilibrium outcome of section 5, the reform at every ρ and its dependence on ϵ and θ included, are identical under the two; only the productivity and leisure parameters and the reading of relative hours differ. That calibration, and the reform at every ρ under both, are reported in the [online appendix](#).

E Calibration – the U.S., France, and the UK

This appendix documents the data and the household heterogeneity behind the calibrations of section 6, whose parameters are in table 3. Each country has three income groups of men aged 25 to 60. The tables report, group by group, the population share γ_i , the taste for leisure X_i and the productivity η_i that the calibration assigns, and the propensity to vote μ_i taken from the data. Under the calibration of section 6 the taste for leisure is common across groups, so the X_i row repeats one number, fixed by the country's average workweek, and the income distribution is carried by η_i alone; the [online appendix](#) describes the alternative, with one X_i per group identified from relative hours.

The parameters in the lower panel of table 3 are common to the three countries. The discount factor β matches the U.S. thirty-year interest rate, the gross interest rate net of average multifactor productivity growth estimated by **mkvss08**, and is imposed on France and the UK. The Frisch elasticity is $\xi = 0.30$, as for Argentina, and the capital share $\alpha = 0.30$.³⁸ Pension spending is for 2020 (**OECDpen**), and ν_t is thirty-year gross population growth from census data and projections, corrected for France as described below; the [online appendix](#) reports the calibration at every intertemporal elasticity of the model's grid.

E.1 The U.S.

For the U.S. we use the CPS March Supplement 2019. The low and medium income groups are cut so that their mean incomes are half the mean and the mean, which is what identifies θ from the replacement rates at those two income levels reported in **OECDpen**;

³⁸Averaging adjustments 2 and 3 of **Gollina1** gives capital income shares close to 0.3 for France and the U.S. and slightly lower for the UK.

the remaining households form the high income group. The propensity to vote is from the U.S. Census, Voting and Registration in the Election of November 2020. To pension spending we add Medicare spending from the 2020 Data Book of the Medicare Payment Advisory Commission, and Medicare enters the calibration of θ as a benefit unrelated to income. France and the UK have no counterpart, since their public health spending is not confined to retirees.³⁹

Table 6 shows the two features that set the U.S. apart. Productivity rises steeply across the groups, from 0.82 in the low group to 2.52 in the high one, and the propensity to vote rises with income, from 0.47 to 0.77, so the political objective weighs high-income households well above their population share. The low group holds 58% of households.

Table 6: Household heterogeneity – the U.S.

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	4.03	4.03	4.03	Average hours worked.
η_i	0.82	1.38	2.52	Income distribution.
μ_i	0.47	0.63	0.77	Voting propensity.

E.2 France

For France we use the 2018 wave of the Luxembourg Income Study. France’s system is fully Bismarckian, so $\theta = 1$ whatever the income cuts, and the groups need not be cut at half-mean and mean income. We cut them at the U.S. percentiles instead, which puts the two countries’ household types in one-to-one correspondence and makes the counterfactuals of section 6 a type-by-type replacement. The propensity to vote is from the Comparative Study of Electoral Systems.⁴⁰ Since the retirement age in France in 2020 was 63.5 years, against 66 in the U.S. and the UK, we take France’s population growth in 2020 to be

$$\frac{\text{Population 2020} \times (30 - 2.5)/30}{\text{Population 1990} + \text{Population 2020} \times 2.5/30},$$

and adjust later years in the same way, with the gap between retirement ages expected to shrink from 2.5 to 2 years.

³⁹**BethencourtG08a1** treat public health care and pensions as political complements, which supports counting Medicare among the transfers to the old that the U.S. electorate votes on.

⁴⁰<https://cses.org/>.

At the same percentiles France’s productivity profile is flatter than the U.S. one (table 7): it runs from 1.12 to 2.56, a ratio of the high to the low group of 2.29 against 3.08 in the U.S. (table 3). The propensity to vote is nearly flat, between 0.81 and 0.86. The common taste for leisure is 7.27 against 4.03 in the U.S., which is how the calibration accounts for France’s shorter workweek.

Table 7: Household heterogeneity – France

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.58	0.18	0.24	Income percentiles.
X_i	7.27	7.27	7.27	Average hours worked.
η_i	1.12	1.58	2.56	Income distribution.
μ_i	0.81	0.86	0.85	Voting propensity.

E.3 The UK

For the UK we use the Harmonised British Household Panel Survey, wave 17, which refers to 2006–07 and is distributed by the UK Data Service.⁴¹ Income groups are cut as for the U.S., at half-mean and mean income. The propensity to vote is from the Comparative Study of Electoral Systems.

The UK’s cuts put fewer households in the low group than the U.S. cuts do, 29% against 58%, and most of them in the medium group (table 8). The ratio of high to low productivity is close to the U.S. one, 3.06 against 3.08, and the propensity to vote lies between the other two countries, rising from 0.61 to 0.74. Table 5 also reports the UK with its households regrouped at the U.S. income percentiles, by a linear fit of the UK’s group means rather than a recount of the survey; the [online appendix](#) documents that grouping.

⁴¹Later waves were collected at least in part during the Great Recession.

Table 8: Household heterogeneity – UK

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	8.26	8.26	8.26	Average hours worked.
η_i	0.96	1.63	2.92	Income distribution.
μ_i	0.61	0.70	0.74	Voting propensity.

E.4 France at the UK’s income groups

Since France’s system is fully Bismarckian, its income groups can also be cut at the UK’s own income percentiles without disturbing the identification of θ , which puts France’s and the UK’s household types in one-to-one correspondence exactly as the U.S. percentiles do above. Table 9 reports France’s household heterogeneity on those groups; the [online appendix](#) imposes them on the UK.⁴² At the UK’s cuts France is the more unequal of the two: the productivity of its top group relative to its bottom group is about 3.3 against the UK’s 3.06, whereas at the U.S. cuts France is well below the U.S. (table 3). The regrouping moves none of France’s aggregates: its political weight of the old and its tax rate are those of table 3.

Table 9: Household heterogeneity – France, UK income groups

Parameter	Income group			Target
	Low	Medium	High	
γ_i	0.29	0.58	0.13	Income percentiles.
X_i	7.20	7.20	7.20	Average hours worked.
η_i	0.94	1.51	3.09	Income distribution.
μ_i	0.81	0.83	0.85	Voting propensity.

F Endogenous pension design under ageing

This appendix reports the one counterfactual of section 7 whose table the paper keeps, the endogenous design under ageing. The counterfactuals of section 6 under general CRRA preferences and under the alternative calibration of the taste for leisure, the same

⁴²Income, hours and group sizes are the same Luxembourg Income Study sample as table 7, regrouped. Voting propensities are only observed at the U.S. cuts; at the UK’s cuts each group carries the population-weighted average of the observed values it overlaps with.

exercise with the UK as the host economy, and the counterfactuals of section 7 that impose France’s characteristics are in the [online appendix](#).

Table 10 reports the acute-ageing counterfactual of section 7 with the design pinned at the U.S. value and with the design chosen, at $\rho \in \{0.5, 1, 2\}$; figure 7.1 collects it with the counterfactuals that impose France’s characteristics. The chosen design rises from 0.738 to 0.816 at $\rho = 0.5$ and to 0.825 and 0.826 at $\rho = 1$ and 2, so the response to ageing hardly depends on the elasticity, and the tax rate ends between 0.1 and 0.3 p.p. above the pinned reading, since a more Bismarckian system dissipates less of its revenue and is worth a larger tax to the electorate. The additional fall in the savings rate is at most 0.4 p.p. of GDP, and the workweek is unchanged to a tenth of an hour.

Table 10: Endogenous design and “acute ageing” in the U.S.

Scenario	CRRA (ρ)	θ (2020)	Tax rate	Savings rate	Avg. workweek (hours)
Baseline	0.5	0.738	14.41%	16.10%	39.45
	1.0	0.738	14.43%	15.41%	39.47
	2.0	0.738	14.44%	14.87%	39.42
Exogenous θ	0.5	0.738	23.01%	−1.91 p.p.	39.00
	1.0	0.738	22.51%	−1.85 p.p.	38.83
	2.0	0.738	21.79%	−1.76 p.p.	38.76
Endogenous θ	0.5	0.816	23.32%	−2.28 p.p.	38.97
	1.0	0.825	22.80%	−2.04 p.p.	38.87
	2.0	0.826	21.92%	−1.81 p.p.	38.84

Note: The deadweight cost $f(\theta_t, \tau_t)$ of equation (17) is quadratic in the implicit tax the flat component levies and scaled by the size of the system, $f = \exp\{-\frac{1}{2}\lambda\tau_t\tilde{V}(1-\theta_t)^2\}$, with λ calibrated per ρ (table 4). Each ρ is separately calibrated and its workweek normalised against its own baseline. The savings rate is savings relative to GDP; the baseline rows report its level and every other row the change against the baseline at the same ρ , in percentage points. The “acute ageing” scenario sets $\nu_t = 1$ throughout.

A cost attached to the design itself rather than to the transfer it performs, $f(\theta) = \phi + (1 - \phi)\theta^p$, is the alternative specification of the [online appendix](#); figure G.1 sets the designs chosen under it beside those of the cost of section 7.2.

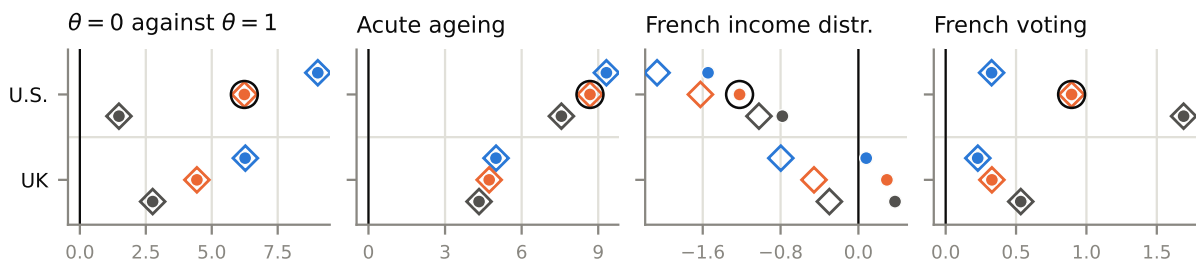
G Robustness

Figure G.1 sets the paper’s main results against the specifications behind them, one panel per result. With the pension design given, the panels are the change in the equilibrium tax

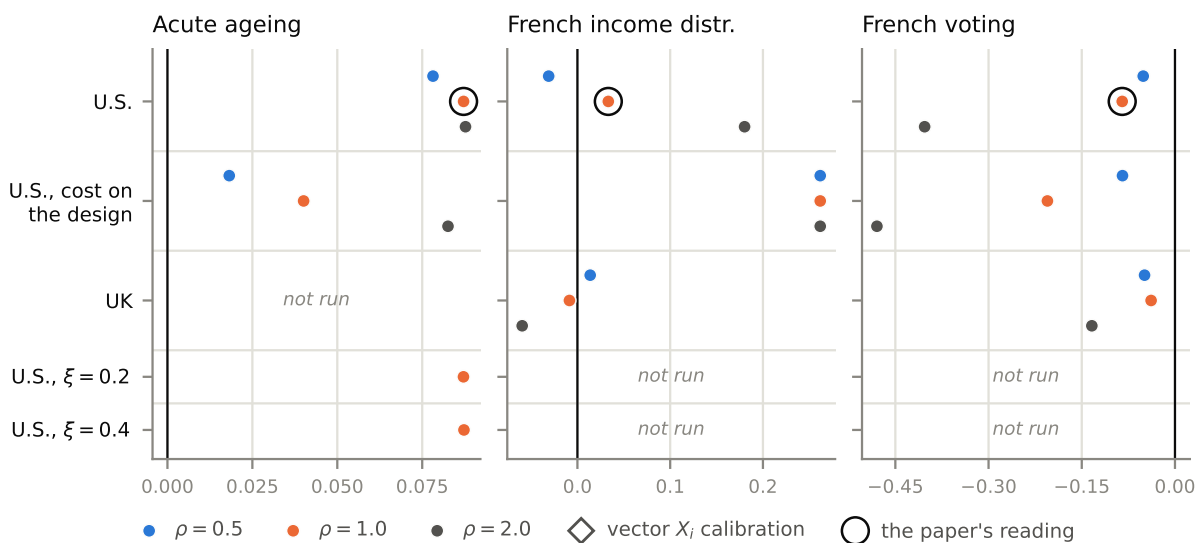
rate from $\theta = 1$ to $\theta = 0$ and its change under acute ageing, France's income distribution and France's voting patterns, in percentage points; with the design chosen, they are the change in the chosen design under the same three counterfactuals. Each counterfactual is read against the host economy's own baseline at the same ρ . Each marker is one specification: its colour is the intertemporal elasticity, its shape the calibration of the taste for leisure, common X or vector X_i ([online appendix](#)), and its row group the host economy. For the chosen design, which is solved under a common X only, further rows hold the alternative cost of the [online appendix](#), the UK as host under France's characteristics and, under acute ageing at $\rho = 1$, the Frisch elasticities $\xi = 0.2$ and 0.4 . The paper's own reading, the U.S. under a common X at $\rho = 1$ with the cost of section 7.2, is highlighted, and a line marks zero: the spread of a panel's markers around the highlighted one is how far the result depends on the specification, and a marker on the other side of the line is a change of sign. The online appendix at championape.github.io/SSDclaude is organized around this figure and links each marker to the table and the row that print its value.

Figure G.1: The main results across specifications

Design given: change in the tax rate, percentage points



Design chosen: change in the design θ in force in 2020



Note: Every value is read in 2020. Table 10 prints the design chosen under acute ageing for the U.S. under a common X . Every other value is printed in the [online appendix](#), with the design given under both calibrations and on both hosts, and with the design chosen.